# SUMS OF TWO SQUARES WITH FEW PRIME FACTORS, II: THE MEAN SQUARE OF THE CUSPIDAL ERROR OF $\theta^2(mz)\theta^2(m'z)$, ALMOST ALL $N$, AND THE SIZE OF THE PETERSSON NORM

RUQING CHEN

ABSTRACT. Let $m, m'$ be coprime odd integers, $\Theta_{m,m'}(z) = \theta^2(mz)\theta^2(m'z)$, $G_{m,m'}$ its cuspidal part and $L = 4mm'$, so that $a_G(N) = T(m, m'; N) - M(m, m'; N)$ is the error term in the number of representations of $N$ by $m(x^2+y^2)+m'(z^2+w^2)$. Part I proved $\|G_{m,m'}\|^2 \ll_\varepsilon (mm')^\varepsilon$ for the Petersson norm and deduced the level of distribution $1/2$ for the weighted sequence $r(n)r(N-n)$ and the bounds $\Omega(n(N-n)) \le 8$, $\max(\Omega(n), \Omega(N-n)) \le 5$ for every large $N \equiv 2 \pmod 4$. Here we prove a mean-square bound $\sum_{N\le Y} |a_G(N)|^2 \le 4\pi e^2 Y(1 + 54Y/L)\|G\|^2$ for every $G \in S_2(\Gamma_0(L))$ by a covering argument; for $G_{m,m'}$ it says that the root mean square of $(T - M)/\sqrt{N}$ is $\ll (mm')^{-1/2+\varepsilon}$, and it gives the level of distribution $1 - \varepsilon$ for the weighted sequence for all but $O(X^{1-\delta})$ integers $N \le X$, and for those $N$ the bounds 5 and 4 (a result which, as an arithmetic statement about almost all $N$, is superseded by the circle method, which gives $N = p + q$ with $p \equiv q \equiv 1 \pmod 4$). We reformulate the uniform bound of Part I in terms of Poincaré series, state the individual bound $|T - M| \ll N^{1/2+\varepsilon}(mm')^{-1/2+\varepsilon}$ suggested by the data, and show that any power saving in $mm'$ for individual $N$ raises the level of distribution for every $N$ beyond $1/2$. Finally we prove the lower bound $\|G_{m,m'}\|^2 \ge \frac{1}{\pi}\big((\log m)^2 + (\log m')^2\big) - O\big(\log(2mm')(\log\log 3mm')^C\big)$ by integrating $|G|^2$ over the parts of the fundamental domain above the four Atkin–Lehner cusps at which $G_{m,m'}$ is transformed into $G_{1,mm'}$, whose first $mm'$ coefficients are $r(n) - M(1, mm'; n)$; so the norm is of the order $(\log mm')^2$ up to the factor $(mm')^\varepsilon$, and the exponent $\varepsilon$ cannot be replaced by 0. Numerically, the regions above all the cusps account for 93–98% of the norm, which is a quadratic form in $\log m$, $\log m'$ plus contributions of the other divisors of $mm'$ rather than a constant times $(\log mm')^2$; there is an exact check at the level 20, and the supremum of $|T - M|/\sqrt{N}$ over $N$ exceeds the root mean square by a bounded factor at every level computed. Finally, replacing the vector sieve of Part I by the two-dimensional sieve of Diamond, Halberstam and Richert, whose functions $F_2, f_2$ we compute, gives $\Omega(n(N-n)) \le 7$ with $\max(\Omega(n), \Omega(N-n)) \le 5$ for every large $N \equiv 2 \pmod 4$ from the same level $1/2$; the two inequalities for $F_2, f_2$ that this needs are certified in interval arithmetic.

## 1. RESULTS

We keep the notation of [5], referred to as Part I. Thus $r(n)$ is the number of representations of $n$ as a sum of two squares, $m, m'$ are coprime odd positive integers,

$$T(m, m'; N) = \sum_{ma+m'b=N} r(a)r(b), \qquad L = 4mm',$$

$$\Theta_{m,m'}(z) = \theta^2(mz)\theta^2(m'z) = \sum_{N\ge 0} T(m, m'; N)e(Nz) \in M_2(\Gamma_0(L)),$$

$\Theta_{m,m'} = E + G$ is the decomposition into the Eisenstein part, whose coefficients are the Siegel main terms $M(m, m'; N)$, and the cuspidal part $G = G_{m,m'} \in S_2(\Gamma_0(L))$, so that

$$a_G(N) = T(m, m'; N) - M(m, m'; N) \qquad \text{for every } N \ge 1.$$

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The Petersson norm is $\|G\|^2 = \int_{\Gamma_0(L)\backslash\mathbb{H}} |G|^2 \,\mathrm{d}x\,\mathrm{d}y$. Lemma 2.1 of Part I, which is Waibel's theorem [11, Theorem 1] for the form $Q_{m,m'} = m(x^2+y^2) + m'(z^2+w^2)$, gives $\|G\| \ll (mm')^{1/4+\varepsilon}$, and the first two versions of Part I deduced from it, through a Hecke operator, the uniform bound $a_G(N) \ll d(N) N^{1/2} (mm')^{1/4+\varepsilon}$ for $(N, mm') = 1$, $v_2(N) \le 1$, the level of distribution 2/5 and the bounds 9 and 6, and conjectured (Conjecture 5.1 there) that the factor $(mm')^{1/4}$ can be removed. Version 3 of Part I, which this paper accompanies, proves that the norm itself is much smaller than Waibel's general bound, by computing the expansion of $\Theta_{m,m'}$ at every cusp of $\Gamma_0(L)$ with Waibel's method, and draws the consequences. We recall what we use.

**Theorem 1.1** (Part I, Theorems 1.2 and 2.3)**.** *Let $m, m'$ be coprime odd positive integers.*

(i) $\|G_{m,m'}\|^2 \ll_\varepsilon (mm')^\varepsilon$.

(ii) *For $N \ge 1$ with $(N, mm') = 1$ and $v_2(N) \le 1$,*

$$T(m, m'; N) = M(m, m'; N) + O_\varepsilon\big(d(N)\, N^{1/2} (mm')^\varepsilon\big);$$

*for cube-free $m, m'$ and every $N$ with $v_2(N) \le 1$ the error term is $O_\varepsilon(d(N)^4 N^{1/2} (mm')^\varepsilon)$ (Part I, Corollary 2.7).*

Part I deduces from (ii) that $r(n)r(N-n)$ has level of distribution 1/2 (Theorem 3.2 there) and, with sieve parameters certified in interval arithmetic, that for every sufficiently large $N \equiv 2 \pmod 4$

$$\#\big\{1 \le n < N :\ n = x^2+y^2,\ N-n = z^2+w^2,\ \Omega(n(N-n)) \le 8,\ \max(\Omega(n), \Omega(N-n)) \le 5\big\} \ge\ 0.23\, \mathfrak{S}_r(N)\, \frac{N}{(\log N)^2}, \qquad \mathfrak{S}_r(N) = \prod_{\substack{p \mid N \\ p > 2}} \frac{p-1}{p-2},$$

(Theorem 1.1 there; versions 1–2 had 9 and 6). The proof of (i) shows that the expansion of $\Theta_{m,m'}$ at every cusp is again a theta series of a form $A(x_1^2+x_2^2) + B(x_3^2+x_4^2)$ with $AB \asymp mm'$, whose coefficients are $O(n^\varepsilon)$ for $n \ll mm'$ because $A\alpha + B\beta = n$ has $O(1)$ solutions, and suggests $\|G_{m,m'}\|^2 \asymp (\log mm')^2$ (Part I, Remark 2.13); numerically, $\|G_{m,m'}\|^2 \approx 0.37\,(\log mm')^2$ (Section 5). Theorem 1.5 below proves the lower bound, and Section 6 decomposes the norm over the cusps.

The first theme of this paper is the average over $N$ of the error term, which is of independent interest and gives a stronger level of distribution for almost all $N$.

**Theorem 1.2** (Mean square)**.** *Let $L \ge 1$, $G \in S_2(\Gamma_0(L))$ and $Y \ge 1$. Then*

$$\sum_{n \le Y} |a_G(n)|^2 \ \le\ 4\pi e^2\, Y \Big(1 + \frac{54\, Y}{L}\Big) \|G\|^2. \tag{1}$$

*In particular, for coprime odd $m, m'$, $G = G_{m,m'}$ and $Y \ge mm'$, by Theorem 1.1(i),*

$$\sum_{N \le Y} \big|T(m, m'; N) - M(m, m'; N)\big|^2 \ \ll_\varepsilon\ Y^2\, (mm')^{-1+\varepsilon}, \tag{2}$$

*and hence, since $\sum_{Y/2 < N \le Y} |a_G(N)|^2/N \ \le\ (2/Y) \sum_{N \le Y} |a_G(N)|^2$, the root mean square of $(T-M)/\sqrt{N}$ over $Y/2 < N \le Y$ is $\ll (mm')^{-1/2+\varepsilon}$.*

The proof of (1) is a classical covering argument and does not use the Petersson formula; the dependence on the level is exactly the factor $Y/L$, i.e. the volume of $\Gamma_0(L)\backslash\mathbb{H}$. The bound (2) is the exact order of magnitude: by the Rankin–Selberg method the mean square of $(T-M)/\sqrt{N}$ is asymptotic to $16\pi^2 \|G\|^2 / \operatorname{vol}(\Gamma_0(L)\backslash\mathbb{H})$, and Table 1 shows the data.

Averaging over $N$ is natural for the arithmetic application, since the sieve of Part I is applied to one $N$ at a time and an exceptional set of $N$ can be tolerated. Let, as in Part I, $N \equiv 2 \pmod 4$, $\mathcal{A} = \{n \le N : \ (n,N)=1\}$, $w(n) = r(n)r(N-n)$, and for coprime square-free $d, e$ with $(de, 2N) = 1$ let $R(d,e) = R(d,e;N)$ be the remainder in

$$\sum_{\substack{n\in\mathcal{A}\\ d\mid n,\ e\mid N-n}} w(n) = X_N\, h(d)h(e) + R(d,e;N)$$

(this is $S(d,e) = M(d,e) + R(d,e)$ of Part I, Section 3; we write $X_N$ for the main term $X$ there, to avoid a clash with the length $X$ of the range of $N$ below).

**Theorem 1.3** (Level of distribution $1-\varepsilon$ for almost all $N$)**.** *Let $0 < \varepsilon < 1/20$, $X \ge X_0(\varepsilon)$ and $D = X^{1-3\varepsilon}$. Then*

$$\sum_{\substack{X/2<N\le X\\ N\equiv 2\ (4)}} \sum_{\substack{de\le D,\ (d,e)=1\\ (de,2N)=1}} \mu^2(de)\, d_3(de)\, |R(d,e;N)| \ \ll_\varepsilon \ X^{2-\varepsilon/2}.$$

*Consequently, for all but $O_\varepsilon(X^{1-\varepsilon/4})$ integers $N \equiv 2 \pmod 4$ in $(X/2, X]$,*

$$\sum_{\substack{de\le N^{1-3\varepsilon}\\ (d,e)=1,\ (de,2N)=1}} \mu^2(de)\, d_3(de)\, |R(d,e;N)| \ \le \ N^{1-\varepsilon/5}. \tag{3}$$

The sieve of Part I, Section 4, uses the level of distribution only through (3); running it at the level $\theta = 19/20$ gives the following.

**Corollary 1.4.** *For all but $O(X^{1-\delta})$ integers $N \equiv 2 \pmod 4$ in $(X/2, X]$, with an absolute $\delta > 0$,*

$$\#\big\{1 \le n < N : \ n = x^2+y^2,\ N-n = z^2+w^2,\ \Omega(n(N-n)) \le 5,\ \max(\Omega(n), \Omega(N-n)) \le 4\big\} \ge \ 0.34\, \mathfrak{S}_r(N)\, \frac{N}{(\log N)^2},$$

*the integers counted being coprime to $N$ and products of distinct primes $\equiv 1 \pmod 4$ exceeding $N^{2/25}$.*

The certified sieve parameters are $\alpha = 2/25$, $\lambda = 1/3$, $\beta_1 = 69/100$ (Section 3). The pairs $(5,4)$ for almost all $N$ and $(8,5)$ for every $N$ should be compared with $(9,6)$ for every $N$, which is what Waibel's bound alone gives (Part I, Remark 4.5). A floating-point scan (Figure 2) shows that the best attainable value of $1/\lambda + 1/\beta_1$ drops below 4 only for $\theta > 0.88$; at $\theta = 19/20$ the scan gives 3.83 (at $\alpha = 0.10$, $\beta_1 = 0.47$, $\lambda = 0.59$, with $1/\lambda + 2/\beta_1 = 5.96$), i.e. the pair $(5,3)$, which we have not certified, for the reason explained next.

Corollary 1.4 must also be compared with what the circle method gives for almost all $N$. By the method of Montgomery and Vaughan [9], carried out with the characters modulo 4 (see [4] for Goldbach's problem with primes in prescribed residue classes), all but $O(X^{1-\delta})$ integers $N \equiv 2 \pmod 4$ up to $X$ are sums $p+q$ of two primes $p \equiv q \equiv 1 \pmod 4$, with $\gg \mathfrak{S}_r(N)N/(\log N)^2$ representations; such primes are sums of two squares, so for almost all $N$ the pair $(1,1)$ holds with the count of Corollary 1.4. As an arithmetic statement about almost all $N$, Corollary 1.4 is therefore far from new. What is new is Theorem 1.3, a level of distribution $1-\varepsilon$ for the weighted sequence $r(n)r(N-n)$, together with the observation that the sieve of Part I extracts $(5,4)$ from it with no further input; the sieve route adds something only for *every* $N$, where the circle method gives nothing (Goldbach's problem) and where Part I gives $(8,5)$. Section 4 formulates

the individual bound that would raise the level to $1-\varepsilon$ for every $N$ and with it give $(5,4)$ for every $N$ (Proposition 4.2).

The third theme is the size of the norm itself. Theorem 1.1(i) is an upper bound. The proof in Part I shows that the expansion of $\Theta_{m,m'}$ at every cusp of $\Gamma_0(L)$ is a theta series of a form $A(x_1^2+x_2^2)+B(x_3^2+x_4^2)$ with $AB \asymp mm'$; at the four cusps which are the images of $\infty$ under the Atkin–Lehner involutions $W_m$, $W_{4m}$, $W_{m'}$, $W_{4m'}$ this form is $x_1^2+x_2^2+mm'(x_3^2+x_4^2)$, whose first $mm'$ representation numbers are the $r(n)$ themselves, and there $G_{m,m'}$ is transformed, exactly, into $G_{1,mm'}$. Integrating $|G|^2$ over the parts of the fundamental domain above these four cusps gives a lower bound.

**Theorem 1.5** (Lower bound for the norm)**.** *Let $m, m'$ be coprime odd positive integers with $mm' > 1$, and let $a(n) = a_{G_{1,mm'}}(n) = T(1,mm';n) - M(1,mm';n)$, so that $a(n) = r(n) - M(1,mm';n)$ for $1 \le n < mm'$.*

(i) $$\|G_{m,m'}\|^2 \;\ge\; \frac{1}{4\pi} \sum_{Q\in\{m,\,4m,\,m',\,4m'\}} \sum_{n\ge 1} \frac{|a(n)|^2}{n}\, e^{-4\pi n/Q}.$$

(ii) *There is an absolute constant $C$ such that*

$$\begin{aligned}\|G_{m,m'}\|^2 &\ge \frac{1}{\pi}\big((\log m)^2+(\log m')^2\big) - C\log(2mm')\,(\log\log 3mm')^C\\ &\ge \frac{1}{2\pi}(\log mm')^2 - C\log(2mm')\,(\log\log 3mm')^C.\end{aligned}$$

Together with Theorem 1.1(i), $(\log mm')^2 \ll \|G_{m,m'}\|^2 \ll_\varepsilon (mm')^\varepsilon$ for $mm' \ge C'$, and the exponent $\varepsilon$ cannot be replaced by 0. The inequality (i) is nearly an equality when $m=1$: the four regions then cover the fraction $5/(2\pi) = 0.80$ of the area of $\Gamma_0(L)\backslash\mathbb{H}$, and numerically they carry 98% of the norm (Table 3). The proof is in Section 6, which also computes, for square-free $m, m'$, the contribution of every cusp and explains the empirical law of Section 5: the norm is not $c(\log mm')^2$ but $\frac{1}{\pi}((\log m)^2+(\log m')^2) + \frac{5}{\pi^2}\log m\log m' + \cdots$, a sum over the divisors of $mm'$.

The fourth theme is the sieve itself. Part I sifts $n$ and $N-n$ by the vector sieve of Brüdern and Fouvry, two linear sieves combined; but $n(N-n)$ is a sieve problem of dimension 2 (Remark 3.3), and the two-dimensional sieve of Diamond, Halberstam and Richert [7, 6], with Richert's weights, has the smaller sieving limit $\beta_2 = 4.2665$. Section 7 computes its sieve functions $F_2, f_2$ (we determine the constants $\alpha_2 = 5.3577$, $\beta_2 = 4.2665$ of [7] independently) and obtains, from the same level of distribution $1/2$ of Part I,

$$\Omega(n(N-n)) \le 7, \qquad \max(\Omega(n), \Omega(N-n)) \le 5$$

for $\ge 0.25\,\mathfrak{S}_r(N)N/(\log N)^2$ integers $n$, for every large $N \equiv 2 \pmod 4$ (Theorem 7.1): $(7,5)$ in place of $(8,5)$. The sieve functions enter through two numerical inequalities, $f_2(9) \ge 0.9995$ and a bound for an integral of $F_2$ (Lemma 7.2), which we certify in interval arithmetic from the constants $\alpha_2, \beta_2$ computed in ball arithmetic by Booker and Browning [2], as Part I does for the linear sieve functions; the theorem is unconditional.

Section 4 restates the uniform bound of Theorem 1.1(ii) in terms of Poincaré series and formulates the individual bound, Section 5 contains the numerical evidence: the data of Part I, an exact check of the whole numerical chain at the level 20, the empirical law $\|G_{m,m'}\|^2 \approx 0.37(\log mm')^2$, which corrects a remark in the first two versions of Part I, and the supremum of $|T-M|/\sqrt{N}$ over $N$; Section 6 proves Theorem 1.5 and decomposes the norm over the cusps; Section 7 contains the two-dimensional sieve; and Section 8 describes the sum over the moduli that a level beyond $1/2$ for every $N$ would require, in the form of a dispersion problem, with numerical evidence that its terms cancel like random signs.

## 2. Proof of Theorem 1.2

Let $G(z)=\sum_{n\ge1}a(n)e(nz)\in S_2(\Gamma_0(L))$ and put $F(z)=y^2|G(z)|^2$, $z=x+iy$. Since $G(\gamma z)=(cz+d)^2G(z)$ and $\operatorname{Im}(\gamma z)=y/|cz+d|^2$ for $\gamma=\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)\in\Gamma_0(L)$, the function $F$ is $\Gamma_0(L)$-invariant, and with the invariant measure $\mathrm{d}\mu=y^{-2}\mathrm{d}x\,\mathrm{d}y$ we have $\|G\|^2=\int_{\mathcal F}F\,\mathrm{d}\mu$ for any fundamental domain $\mathcal F$ of $\Gamma_0(L)$.

Let $y_0=1/(4\pi Y)$ and $R=\{x+iy:\ 0\le x<1,\ y_0\le y\le 2y_0\}$ (half-open in $x$, so that each orbit of the translations $z\mapsto z+b$ meets $R$ at most once). By Parseval, $\int_0^1|G(x+iy)|^2\,\mathrm{d}x=\sum_n|a(n)|^2e^{-4\pi ny}$, hence

$$\int_R|G|^2\,\mathrm{d}x\,\mathrm{d}y=\sum_n|a(n)|^2\int_{y_0}^{2y_0}e^{-4\pi ny}\,\mathrm{d}y\ \ge\ \frac{e^{-2}}{4\pi Y}\sum_{n\le Y}|a(n)|^2,$$

because $4\pi ny\le 8\pi Yy_0=2$ for $n\le Y$, $y\le 2y_0$. On the other hand $\int_R|G|^2\,\mathrm{d}x\,\mathrm{d}y=\int_R F\,\mathrm{d}\mu$, and

$$\int_R F\,\mathrm{d}\mu=\sum_{\gamma\in\Gamma_0(L)/\{\pm1\}}\int_{R\cap\gamma\mathcal F}F\,\mathrm{d}\mu=\int_{\mathcal F}F(w)\,\#\{\gamma:\ \gamma w\in R\}\,\mathrm{d}\mu(w)\ \le\ \mu_R\,\|G\|^2,$$

where $\mu_R=\sup_{z\in R}\#\{\gamma\in\Gamma_0(L)/\{\pm1\}:\ \gamma z\in R\}$ (if $\#\{\gamma:\gamma w\in R\}\ne0$ for some $w\in\mathcal F$, pick $\gamma_0$ with $z=\gamma_0 w\in R$; then $\gamma\mapsto\gamma\gamma_0^{-1}$ is a bijection from $\{\gamma:\gamma w\in R\}$ onto $\{\gamma:\gamma z\in R\}$, so the count at $w$ is at most $\mu_R$). It remains to bound $\mu_R$. Fix $z=x+iy\in R$ and let $\gamma=\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)\in\Gamma_0(L)$ with $\gamma z\in R$.

- If $c=0$, then $\gamma=\pm\left(\begin{smallmatrix}1&b\\0&1\end{smallmatrix}\right)$ and $\gamma z=z+b$, so $x+b\in[0,1)$ forces $b=0$: one element.
- If $c\ne0$, we may take $c>0$, since $\gamma$ and $-\gamma$ are the same element of $\Gamma_0(L)/\{\pm1\}$. Then $L\mid c$, and $\operatorname{Im}(\gamma z)=y/|cz+d|^2\in[y_0,2y_0]$ together with $y\in[y_0,2y_0]$ gives $|cz+d|^2\in[1/2,2]$. Since $|cz+d|^2=(cx+d)^2+c^2y^2$, this forces $c^2y^2\le2$, hence $c\le\sqrt2/y_0=4\sqrt2\,\pi Y$, and $|cx+d|\le\sqrt2$. The number of positive multiples $c$ of $L$ with $c\le4\sqrt2\pi Y$ is $\lfloor4\sqrt2\pi Y/L\rfloor\le4\sqrt2\pi\,Y/L$; for each $c$ there are at most 3 integers $d$ with $|cx+d|\le\sqrt2$; and for each pair $(c,d)$ the elements of $\Gamma_0(L)$ with bottom row $(c,d)$ are $\left(\begin{smallmatrix}1&k\\0&1\end{smallmatrix}\right)\gamma_0$, $k\in\mathbb Z$, whose images $\gamma_0z+k$ have real part in $[0,1)$ for exactly one $k$. This gives at most $4\sqrt2\pi\cdot3\,Y/L=12\sqrt2\pi\,Y/L<54\,Y/L$ elements.

Hence $\mu_R\le1+54Y/L$, and

$$\sum_{n\le Y}|a(n)|^2\le4\pi e^2Y\int_R F\,\mathrm{d}\mu\le4\pi e^2\,Y\Big(1+\frac{54Y}{L}\Big)\|G\|^2,$$

which is (1). For $G=G_{m,m'}$, $L=4mm'$ and $Y\ge mm'$ the bracket is $\le1+54Y/(4mm')\le15Y/(mm')$, and Theorem 1.1(i) gives (2). □

*Remark* 2.1. The argument is the classical one for bounding $\sum_{n\le Y}|a(n)|^2$ by the norm (compare [10, Chapter 5]); its only input is the invariance of $y^2|G|^2$. The strip $y_0\le y\le2y_0$ with $y_0=1/(4\pi Y)$ is chosen so that the exponential factor in Parseval's identity stays at $e^{-2}$ for all $n\le Y$. For $Y\ge L$ the factor $Y^2/L$ in (1) is of the correct order of magnitude: by the Rankin–Selberg method, $\sum_{n\le Y}|a(n)|^2/n\sim16\pi^2\|G\|^2Y/\operatorname{vol}(\Gamma_0(L)\backslash\mathbb H)$ with $\operatorname{vol}=\frac{\pi}{3}L\prod_{p\mid L}(1+1/p)$ (measure $\mathrm{d}\mu$), so that $\sum_{n\le Y}|a(n)|^2\sim8\pi^2\|G\|^2Y^2/\operatorname{vol}$ for fixed $G$ as $Y\to\infty$. Theorem 1.2 makes this uniform in the level, at the cost of the constant.

## 3. Proof of Theorem 1.3 and Corollary 1.4

3.1. **The remainder on average.** We follow the proof of Theorem 3.2 of Part I, replacing the uniform bound for $|T-M|$ by Theorem 1.2 after an application of Cauchy–Schwarz over $N$.

Recall from Part I, (16), that for $N \equiv 2 \pmod 4$ and coprime square-free $d, e$ with $(de, 2N) = 1$,

$$(4)\quad R(d,e;N) = \sum_{t \mid N^\flat} \mu(t) \sum_{m \mid d^2} \sum_{m' \mid e^2} c_d(m) c_e(m') \sum_{\substack{t_1 \mid t^2 \\ t \mid t_1}} c_t(t_1) \big(a_G(N) - a_G(N/2)\big), \qquad G = G_{t_1 m, m'},$$

where $N^\flat$ is the product of the odd primes dividing $N$, $m = ds$, $m' = es'$ with $s \mid d$, $s' \mid e$, and likewise $t_1 = tu$ with $u \mid t$ (the coefficients $c_d(m)$, $c_t(t_1)$ of Part I, (15), vanish unless $m = ds$ with $s \mid d$, resp. $t_1 = tu$ with $u \mid t$; in particular $t \mid t_1$); the parameters $t_1 m, m'$ of the form $Q_{t_1 m, m'}$ are coprime, odd and cube-free, $\tilde{m}\tilde{m}' := t_1 m m' = t_1 s s' \cdot de$, and the coefficients $c$ satisfy $|c_d(m) c_e(m') c_t(t_1)| \le 3^{\omega(d)+\omega(e)+\omega(t)}$. (For the reader's convenience: $c_d$ is the multiplicative coefficient of Part I, (15), with $c_p(p) = 2$, $c_p(p^2) = -1$ for a split prime $p$ and $c_q(q) = 0$, $c_q(q^2) = 1$ for an inert prime $q$; it encodes the identities $r(n) = 2r(n/p) - r(n/p^2)$ and $r(n) = r(n/q^2)$, which turn the condition $d \mid n$ into a combination of theta series $\theta^2(mz)$. The sum over $t$ is the Möbius inversion of the condition $(n, N) = 1$, and $a_G(N) - a_G(N/2)$ selects the odd $n$.) Let $\eta = \varepsilon/10$. For $N \le X$ with $X \ge X_0(\varepsilon)$ we have, as in (F1)–(F3) of Part I: $d_3(de)\, 3^{\omega(d)+\omega(e)+\omega(t)}\, 4^{\omega(N)} d(N)^2 \le X^\eta$; for fixed $(d, e, N)$ the number of tuples $(s, s', t, t_1)$ is at most $X^\eta$; and $\sum_{k \le D} d(k)/k \le X^\eta$, $\sum_{k \le D} d(k) k^{-1/2} \le D^{1/2} X^\eta$. We also use the two bounds for a single term,

$$(5)\qquad |a_G(N')| \le T(\tilde{m}, \tilde{m}'; N') + M(\tilde{m}, \tilde{m}'; N') \ll X^{2\eta}\Big(\frac{X}{\tilde{m}\tilde{m}'} + 1\Big) \qquad (N' \le X),$$

$$(6)\qquad \sum_{n \le X} |a_G(n)| \le X^{1/2} \Big(\sum_{n \le X} |a_G(n)|^2\Big)^{1/2} \ll X\Big(1 + \frac{X}{L}\Big)^{1/2} L^\eta,$$

where $G = G_{\tilde{m}, \tilde{m}'}$ and $L = 4\tilde{m}\tilde{m}'$, the first being (21) of Part I, the second Cauchy–Schwarz, (1) and Theorem 1.1(i); for $L \le X$ the right-hand side of (6) is $\ll X^{3/2} (\tilde{m}\tilde{m}')^{-1/2+\eta}$. Let $\Sigma$ denote the double sum in Theorem 1.3. Inserting (4) and splitting the tuples according to the size of $t_1 s s'$:

*Tuples with $t_1 s s' > X^\varepsilon$.* For each $N$ and each tuple, (5) with $\tilde{m}\tilde{m}' > X^\varepsilon de$ gives $|a_G(N')| \ll X^{2\eta}(X^{1-\varepsilon}/(de) + 1)$. Summing over the $\le X^\eta$ tuples, over $N \le X$ and over $de \le D$ with the weights $d_3(de) \cdot 3^\omega \le X^\eta$, these tuples contribute

$$\ll X^{1+4\eta} \sum_{k \le D} d(k) \Big(\frac{X^{1-\varepsilon}}{k} + 1\Big) \ll X^{2-\varepsilon+5\eta} + X^{1+5\eta} D \ll X^{2-\varepsilon/2}.$$

*Tuples with $t_1 s s' \le X^\varepsilon$.* Now $t \le t_1 \le X^\varepsilon$ (as $t \mid t_1$), so $t$ runs over at most $X^\varepsilon$ values, $t_1$ over at most $d(t^2) \le X^\eta$ values, and $(s, s')$ over at most $d(d) d(e) \le X^\eta$ values; the forms $G = G_{t_1 m, m'}$ have level $L = 4 t_1 s s' de \le 4 X^\varepsilon D \le X$, so that the right-hand side of (6) is $\ll X^{3/2} (\tilde{m}\tilde{m}')^{-1/2+\eta}$. For a fixed tuple and fixed $(d, e)$, we drop the condition $t \mid N$ except for the count: the number of $N \le X$ with $t \mid N$ is $\le X/t$, and Cauchy–Schwarz gives

$$\begin{aligned}\sum_{\substack{N \le X \\ t \mid N}} \big(|a_G(N)| + |a_G(N/2)|\big) &\le 2\Big(\frac{X}{t}\Big)^{1/2} \Big(\sum_{n \le X} |a_G(n)|^2\Big)^{1/2} \\ &\ll \Big(\frac{X}{t}\Big)^{1/2} X\, (\tilde{m}\tilde{m}')^{-1/2+\eta} \le X^{3/2+\eta}\, t^{-1/2} (de)^{-1/2},\end{aligned}$$

using (1) in the form $\sum_{n \le X} |a_G(n)|^2 \ll X^2 \|G\|^2 / L$ and $(\tilde{m}\tilde{m}')^{-1/2+\eta} \le X^\eta (de)^{-1/2}$. Summing over $t \le X^\varepsilon$ ($\sum_t t^{-1/2} \le 2X^{\varepsilon/2}$), over the other tuple variables ($\le X^{2\eta}$), and over $de \le D$ with

the weights $\le X^{\eta}$,

$$\ll X^{3/2+4\eta+\varepsilon/2}\sum_{k\le D} d(k)k^{-1/2} \ll X^{3/2+5\eta+\varepsilon/2}D^{1/2} = X^{2-\varepsilon+5\eta} = X^{2-\varepsilon/2},$$

since $D^{1/2} = X^{1/2-3\varepsilon/2}$. Both contributions are $\ll X^{2-\varepsilon/2}$, which proves the first assertion. (The margin is small but positive: the choice $D = X^{1-3\varepsilon}$ saves $X^{-3\varepsilon/2}$ in $D^{1/2}$, of which the sum over $t$ costs $X^{\varepsilon/2}$ and the divisor-type factors cost $X^{5\eta} = X^{\varepsilon/2}$; this is why $\eta = \varepsilon/10$.)

For the second, let $S(N)$ be the inner sum with $D = X^{1-3\varepsilon}$. The number of $N \in (X/2, X]$, $N \equiv 2 \pmod 4$, with $S(N) > X^{1-\varepsilon/4}$ is at most $\Sigma/X^{1-\varepsilon/4} \ll X^{1-\varepsilon/4}$. For the remaining $N$, since $N^{1-3\varepsilon} \le D$ and $X^{1-\varepsilon/4} \le (2N)^{1-\varepsilon/4} \le N^{1-\varepsilon/5}$ for large $X$, (3) holds. □

*Remark* 3.1. The exponent 1 is the balance $X^{3/2}D^{1/2} = X^2$ between the mean-square bound $X^{3/2}(de)^{-1/2}$ for the sum over $N$ of one remainder and the number $D$ of moduli; with a norm bound $\|G_{m,m'}\| \ll (mm')^{\beta+\varepsilon}$ in place of Theorem 1.1(i) one gets the level $1/(2(1+\beta))$ for every $N$ (Part I, Remark 3.3) and $1/(1+2\beta)$ for almost all $N$; Waibel's $\beta = 1/4$ gives $2/5$ and $2/3$, and $\beta = 0$ gives $1/2$ and 1. The exceptional set $X^{1-\varepsilon/4}$ comes from Markov's inequality applied to a first moment of $S(N)$, which in turn rests on the second moment of $a_G(N)$; a bound for a higher moment would shrink it, and we do not pursue this.

3.2. **The sieve at the level** 19/20**.** Proposition 4.3 of Part I and its proof use the level of distribution only through the bound (3) (there with 2/5); they hold verbatim, for a given $N$, with $\theta = 2/5$ replaced by any $\theta \in (0,1)$ for which (3) holds with $N^{\theta-\varepsilon}$ in place of $N^{1-3\varepsilon}$, provided $\alpha < \beta_1 < \theta$. Lemmas 4.1 and 4.2 of Part I do not involve the level at all. Hence, for every $N$ satisfying (3) with $3\varepsilon < 1/20$, the conclusion of Proposition 4.3 holds with $\theta = 19/20$, and the proof of Theorem 1.1 of Part I goes through with the parameters certified for $\theta = 19/20$. The script `certify_modular.py` of Part I, extended by the argument `19/20` in the repository of this paper, certifies in interval arithmetic

$$\theta = \tfrac{19}{20}, \qquad \alpha = \tfrac{2}{25}, \qquad \lambda = \tfrac{1}{3}, \qquad \beta_1 = \tfrac{69}{100},$$
$$\tfrac{1}{\lambda} + \tfrac{2}{\beta_1} = \tfrac{407}{69} < 6, \qquad \tfrac{1}{\lambda} + \tfrac{1}{\beta_1} = \tfrac{307}{69} < 5, \qquad L - \lambda R \ge 0.085058$$

(with $L \ge 0.999484$, $R \le 2.743278$, $\theta_A = 0.4740$; the output files are listed at the end of the paper). By Lemma 4.2 of Part I this gives $\Omega(n(N-n)) \le 5$ and $\max(\Omega(n), \Omega(N-n)) \le 4$, and the weight of each $n$ counted is $w(n) = 16 \cdot 2^{\Omega(n(N-n))} \le 2^9$. As in the proof of Theorem 1.1 of Part I, the number of integers counted is at least

$$\big(0.085058 + o(1)\big)\, 2^{-9} \cdot 8\, \mathfrak{S}_r(N)\, c_{2h}\, \frac{N}{\alpha^2 (\log N)^2} \;\ge\; 0.34\, \mathfrak{S}_r(N) \frac{N}{(\log N)^2},$$

since $0.085058 \cdot 2^{-9} \cdot 8 \cdot 1.66485 \cdot (25/2)^2 = 0.3457\ldots$. The exceptional set is that of Theorem 1.3 with, say, $\varepsilon = 1/80$, so $\delta = 1/320$ is admissible. □

*Remark* 3.2. The same sieve gives $(8,5)$ at the level 1/2 (Part I), $(6,4)$ at 2/3 (certified: $\alpha = 3/50$, $\lambda = 1/3$, $\beta_1 = 51/100$, $L - \lambda R \ge 0.036741$) and $(5,4)$ at 19/20. The floating-point scan of Figure 2 shows that $1/\lambda + 2/\beta_1$ drops below 6 only for $\theta > 0.83$, and that the best attainable value of $1/\lambda + 1/\beta_1$ drops below 4 for $\theta > 0.88$: at $\theta = 19/20$ the minimum is 3.83, attained at $\alpha = 0.10$, $\beta_1 = 0.47$, $\lambda = 0.59$, where $1/\lambda + 2/\beta_1 = 5.96$, so that the Richert weights give $(5,3)$ for almost all $N$ in the floating-point model, with the sifting parameter $\alpha$ at the edge of the scanned range. We have not certified this set, because Corollary 1.4 is in any case superseded for almost all $N$ by the circle method (Section 1); it is an "almost all $N$" statement with a power-saving exceptional set and says nothing about any individual $N$, which is where the sieve route matters (Section 4.1).

*Remark* 3.3 (The vector sieve is not the natural sieve here). The sieve of Part I treats the two sides $n$ and $N-n$ by the vector sieve of Brüdern and Fouvry, with two linear components of levels $N^{\theta_A}$ and $N^{\theta-\theta_A}$ and the lower bound $L = f_1(s_A)F_1(s_B) + F_1(s_A)f_1(s_B) - F_1(s_A)F_1(s_B)$. For $s_A, s_B \le 3$ one has $f_1/F_1(s) = \log(s-1)$, so $L > 0$ if and only if $(s_A-1)(s_B-1) > e$, and with $s = s_A + s_B$ fixed this requires $s > 2 + 2\sqrt{e} = 5.297\ldots$, attained at the symmetric split. The product sequence $n(N-n)$, however, is a sieve problem of dimension 2 in the classical sense: under the weight $r(n)r(N-n)$ the density of $p \mid n(N-n)$ is $2h(p)$, $\sum_{p<z} 2h(p) = 2\log\log z + O(1)$, the condition $d \mid n(N-n)$ with $(n,N) = 1$ splits uniquely as $d = d_1 d_2$, $d_1 \mid n$, $d_2 \mid N-n$, and the remainder for the modulus $d$ is $\sum_{d_1 d_2 = d} R(d_1, d_2)$, which is exactly what Theorem 3.2 of Part I and Theorem 1.3 bound (this is the reason for the weight $d_3(de)$ there). The two-dimensional sieve of Diamond, Halberstam and Richert [6] applies to it directly, with the joint level $d \le N^{\theta}$ and the sieving limit $\beta_2 = 4.2665\ldots$ in place of 5.297; its upper and lower functions $F_2, f_2$ should be used in place of the products of $F_1, f_1$ both in $L$ and in the weight term $R$. Section 7 carries this out: with the sieve functions $F_2, f_2$ enclosed in interval arithmetic, the two-dimensional sieve gives $(7,5)$ in place of $(8,5)$ for every $N$ at the level $1/2$ (Theorem 7.1), $(6,4)$ at $2/3$ with a larger margin, and $(5,3)$ at $19/20$; the bound for $\max(\Omega(n), \Omega(N-n))$ comes from the same Richert weights, which see the two sides separately. For comparison, for the unweighted sequence $n(N-n)$, which has level of distribution 1 trivially, the two-dimensional weighted sieve gives $N = P_2 + P_3$ for every large even $N$ (Wang Yuan, 1957).

## 4. The uniform bound and the Poincaré series

Let $L = 4mm'$ and let $P_N \in S_2(\Gamma_0(L))$, $N \ge 1$, be the Riesz representative of the $N$-th coefficient: the unique cusp form with

$$\langle f, P_N \rangle = \frac{a_f(N)}{4\pi N} \qquad \text{for all } f \in S_2(\Gamma_0(L)), \tag{7}$$

i.e. $P_N = (4\pi N)^{-1} \sum_{f \in \mathcal{B}} \overline{a_f(N)}\, f$ for an orthonormal basis $\mathcal{B}$. (This is the $N$-th Poincaré series; for weight 2 the series itself converges only conditionally, which is why we define $P_N$ by (7).) By (11) of Part I,

$$\|P_N\|^2 = \frac{a_{P_N}(N)}{4\pi N} = \frac{1}{(4\pi N)^2} \sum_{f \in \mathcal{B}} |a_f(N)|^2 = \frac{\Delta_L(N,N)}{4\pi N},$$

where $\Delta_L(N,N)$ is the Petersson density. Hence, for $G = G_{m,m'}$,

$$|a_G(N)|^2 = (4\pi N)^2 |\langle G, P_N \rangle|^2 = 4\pi N\, \Delta_L(N,N)\, \|G\|^2 \cos^2 \angle(G, P_N), \tag{8}$$

where $\cos\angle(G,P_N) = |\langle G, P_N\rangle| / (\|G\|\,\|P_N\|)$. The identity separates the three quantities that enter any bound for $a_G(N)$: the norm $\|G\|^2$, which by Theorem 1.1(i) is $\ll L^{\varepsilon}$; the Petersson density, which is $\le C'$ for $N \le L^{1/2}$ (Part I, Lemma 2.2, Step 4) and $1 + O(N^{1/2+\varepsilon}/L)$ in general; and the alignment $\cos^2\angle(G,P_N) \le 1$. Theorem 1.1(ii) is obtained from (8) by bounding the alignment trivially: after the coprime part of $N$ is moved into a Hecke operator (Part I, Lemma 2.2), only $P_{n_2}$ with $n_2 \in \{1,2\}$ occurs, where the density is bounded, and the norm does the rest. Theorem 1.2 says what the alignment is on average: inserting (8) into (1) and restricting to $N \in (Y/2, Y]$,

$$\frac{1}{Y/2} \sum_{Y/2 < N \le Y} \Delta_L(N,N) \cos^2 \angle(G, P_N) \;\le\; 4e^2\Big(\frac{1}{Y} + \frac{54}{L}\Big) \;\ll\; \frac{1}{L} \qquad (Y \ge L), \tag{9}$$

the value of $\cos^2$ for a unit vector in general position in the $\asymp L$-dimensional space $S_2(\Gamma_0(L))$: on average over $N$, the theta series is no more aligned with the Poincaré series than a random cusp form of level $L$ would be, and Theorem 1.1(ii) says that no single $N$ is aligned by more than

$N^\varepsilon L/\|G\|^2 \gg L^{1-\varepsilon}$ times the average. Expanding in an orthogonal basis $\mathcal{B}^*$ of Hecke eigenforms as in Part I, Section 5.5,

$$\langle G, P_N\rangle = \sum_{\varphi\in\mathcal{B}^*} \frac{\langle G,\varphi\rangle\langle\varphi,P_N\rangle}{\|\varphi\|^2} = \frac{1}{4\pi N}\sum_{\varphi\in\mathcal{B}^*} \frac{\langle\Theta,\varphi\rangle\, a_\varphi(N)}{\|\varphi\|^2},$$

so that Theorem 1.1(ii) is the bound $\ll N^{1/2+\varepsilon}$ for this spectral sum, uniformly in $m, m'$; the first two versions of Part I asked for a proof through the cancellation between the coefficients $\langle\Theta,\varphi\rangle$ and $a_\varphi(N)$. The proof in Part I does not see this cancellation: it shows instead that $\sum_\varphi |\langle\Theta,\varphi\rangle|^2/\|\varphi\|^2 = \|G\|^2$ is itself small, i.e. that the theta series has a small cuspidal projection to begin with, and the spectral sum is then bounded by Cauchy–Schwarz.

4.1. **The individual bound.** Theorem 1.2 and the data (Section 5.4) say that the cancellation which the proof does not see is there for most $N$, and the sieve for every $N$ would profit from it for each $N$. We state the strongest form suggested by the data.

*Conjecture* 4.1. For coprime odd $m, m'$ and $N \geq 1$ with $(N, mm') = 1$ and $v_2(N) \leq 1$,

$$|T(m,m';N) - M(m,m';N)| = |a_{G_{m,m'}}(N)| \ \ll_\varepsilon \ N^{1/2+\varepsilon}(mm')^{-1/2+\varepsilon}.$$

In the terms of (8): $\Delta_L(N,N)\cos^2\angle(G,P_N) \ll N^\varepsilon L^{-1+\varepsilon}/\|G\|^2$, i.e. no Poincaré series $P_N$ is aligned with $G$ by more than $N^\varepsilon L^\varepsilon$ times the average (9).

Conjecture 4.1 is the mean-square bound (2) applied to each $N$ separately; it is consistent with the trivial lower bound $|a_G(N)| \geq \operatorname{dist}(M, 8\mathbb{Z})$ for $N \ll L$, where $M \asymp N/(mm') \ll 1$, and with the data of Section 5.4, where the supremum over $10^6$ to $4\cdot 10^6$ values of $N$ exceeds the root mean square by a factor between 5 and 10 at every level computed. We do not know how to prove any power saving in $mm'$ for an individual $N$. In the spectral form the question is a twisted first moment $\sum_\varphi \langle\Theta,\varphi\rangle a_\varphi(N)/\|\varphi\|^2$ over the forms of level $L$ with the "long" twist $N \geq L$ (the sieve at the level $\theta$ uses $N \approx L^{1/\theta}$), outside the range $N \leq L^{1-\delta}$ in which the Petersson formula evaluates such moments. Any saving would raise the level for every $N$:

**Proposition 4.2.** *Let $0 \leq \eta \leq \frac12$ and suppose that $|a_{G_{m,m'}}(N)| \ll_\varepsilon N^{1/2+\varepsilon}(mm')^{-\eta+\varepsilon}$ for all coprime odd $m, m'$ and all $N$ with $(N, mm') = 1$, $v_2(N) \leq 1$. Then Theorem 3.2 of Part I holds for every $N \equiv 2 \pmod 4$ with $D = N^{\theta-\varepsilon}$, $\theta = 1/(2(1-\eta))$, i.e. $r(n)r(N-n)$ has level of distribution $1/(2(1-\eta))$. In particular, Conjecture 4.1 gives the level $1-\varepsilon$ for every $N$, and then the sieve of Section 3 gives, for every sufficiently large $N \equiv 2 \pmod 4$, $\Omega(n(N-n)) \leq 5$ and $\max(\Omega(n), \Omega(N-n)) \leq 4$ for at least $0.34\,\mathfrak{S}_r(N)N/(\log N)^2$ integers $n$.*

*Proof.* We follow the proof of Theorem 3.2 of Part I and only change the bound for one remainder. Each term of $R(d,e)$ is $\pm c\, a_G(N')$ for a form $G = G_{\tilde m,\tilde m'}$ with $\tilde m\tilde m' = t_1 ss'\cdot de$, where $t \mid N^\flat$, $t_1 \mid t^2$, and $N' \in \{N, N/2\}$ may have common factors with $\tilde m\tilde m'$ at the primes of $t$ only. Lemma 2.5 of Part I reduces $a_G(N')$ to $\leq 4^{\omega(N)}d(N)$ terms $a_{G_{m_j,m_j'}}(N_j)$ with $(N_j, m_jm_j') = 1$, $N_j \mid N'$ and $m_jm_j' \geq \tilde m\tilde m'/t_1^2 \geq de/t_1^2$ (each reduction step at a prime $p$ divides the product of the coefficients by at most $p^2$, and only the primes of $t_1$ are involved). Hence, by the hypothesis with $\varepsilon/20$ in place of $\varepsilon$,

$$|a_G(N')| \ll 4^{\omega(N)}d(N)\, N^{1/2+\varepsilon/20}\,(de)^{-\eta}\, t_1^{2\eta}\,(mm')^{\varepsilon/20}.$$

Put $\eta' = \varepsilon/20$. For the terms with $t_1ss' \leq N^\varepsilon$ this is $\ll N^{1/2+3\eta'+2\varepsilon\eta}(de)^{-\eta}$ by (F1) of Part I, and their total is

$$\ll N^{1/2+5\eta'+\varepsilon}\sum_{k\leq D} d(k)k^{-\eta} \ll N^{1/2+6\eta'+\varepsilon}D^{1-\eta} = N^{1/2+6\eta'+\varepsilon}N^{1/2-\varepsilon(1-\eta)} \leq N^{1-\varepsilon/2+6\eta'},$$

since $D^{1-\eta} = N^{(\theta-\varepsilon)(1-\eta)} = N^{1/2-\varepsilon(1-\eta)}$ and $1-\eta \geq \frac{1}{2}$. The terms with $t_1 ss' > N^{\varepsilon}$ are bounded by the trivial bound (5) exactly as in Part I, by $N^{1-\varepsilon+5\eta'} + N^{\theta+5\eta'}$. With $\theta \leq 1-\varepsilon$ (replace $\varepsilon$ by $\min(\varepsilon, 1-\theta)$ if necessary) the total is $\ll N^{1-\varepsilon/2+6\eta'} \leq N^{1-\varepsilon/5}$, which is the conclusion of Theorem 3.2 of Part I with $N^{1-\varepsilon/5}$ in place of $N^{1-\varepsilon/3}$; Proposition 4.3 of Part I needs only a power saving. The last assertion is Section 3, where the parameters at the level 19/20 were certified for any $N$ satisfying (3). □

The exponent $\theta = 1/(2(1-\eta))$ interpolates between the level 1/2 of Part I ($\eta = 0$) and the level 1 ($\eta = \frac{1}{2}$); the level 2/3, which gives $(6, 4)$ for every $N$ with the certified parameters of Section 3, needs only $\eta = \frac{1}{4}$, i.e. a saving of $(mm')^{-1/4}$ over Theorem 1.1(ii), half of what the mean square (2) gives on average.

## 5. Numerical evidence

5.1. **The mean square and the norm.** The script `mean_square_check.py` computes, with the tables and the FFT convolution of `sup_error_scan.py` of Part I, the sums $S_2(Y) = \sum_{N \in \mathcal{N}_Y} |a_G(N)|^2$ over the admissible set

$$\mathcal{N}_Y = \{N \leq Y : \ (N, mm') = 1, \ v_2(N) \leq 3\},$$

on which the main term of that script is exact, for $Y = 2^{21}$ and $Y = 2^{20}$, and the estimate

$$\|G\|^2_{\mathrm{est}} = \frac{\mathrm{vol}(\Gamma_0(L)\backslash\mathbb{H})}{16\pi^2} \cdot \frac{1}{\#\mathcal{N}_Y} \sum_{N \in \mathcal{N}_Y} \frac{|a_G(N)|^2}{N}$$

suggested by the Rankin–Selberg asymptotic of Remark 2.1. Replacing $1/Y$ by $1/\#\mathcal{N}_Y$ assumes that the mean square of $|a_G(N)|^2/N$ over the excluded $N$ (those with $(N, mm') > 1$ or $v_2(N) \geq 4$) is the same as over $\mathcal{N}_Y$; this is immaterial when $mm'$ has only large prime factors, but introduces a bias of unknown sign for $(1, 5)$, $(9, 1)$, $(3, 5)$ and $(1, 15015)$, where 25%, 38%, 50% and 64% of the $N$ are excluded. Table 1 shows the result at $Y = 2^{21}$, and Figure 1 the dependence on $Y$ and on $mm'$.

| $(m, m')$ | $mm'$ | $\#\mathcal{N}_Y$ | $S_2(Y)/S_2(Y/2)$ | $\|G\|^2_{\mathrm{est}}$ | $\|G\|^2_{\mathrm{est}}/(mm')^{1/2}$ | $\|G\|^2_{\mathrm{est}}/(\log mm')^2$ |
|---|---|---|---|---|---|---|
| $(1, 5)$ | 5 | 1572864 | 4.001 | 0.74 | 0.33 | 0.29 |
| $(9, 1)$ | 9 | 1310720 | 4.001 | 1.47 | 0.49 | 0.30 |
| $(3, 5)$ | 15 | 1048576 | 4.001 | 3.44 | 0.89 | 0.47 |
| $(1, 101)$ | 101 | 1946614 | 4.000 | 7.95 | 0.79 | 0.373 |
| $(13, 29)$ | 377 | 1752262 | 4.001 | 13.09 | 0.67 | 0.372 |
| $(1, 1009)$ | 1009 | 1964131 | 3.976 | 17.96 | 0.57 | 0.375 |
| $(61, 67)$ | 4087 | 1904986 | 3.961 | 24.47 | 0.38 | 0.354 |
| $(101, 103)$ | 10403 | 1927715 | 3.948 | 29.81 | 0.29 | 0.348 |
| $(1, 15015)$ | 15015 | 754220 | 4.359 | 36.62 | 0.30 | 0.396 |
| $(1009, 1013)$ | 1022117 | 1962192 | 4.559 | 46.87 | 0.046 | 0.244 |

Table 1. Mean square of the cuspidal error over the admissible $N \leq Y = 2^{21}$ (`results/mean_square_check_x21.txt`). The ratio $S_2(Y)/S_2(Y/2) \approx 4$ is the growth $Y^2$ of Theorem 1.2. For the first three pairs and for $(1, 15015)$ a large proportion of the $N$ (25% to 64%) is excluded by the coprimality condition, and for the last pair $Y < L$, so that neither (1) in its $Y^2/L$ form nor the Rankin–Selberg asymptotic applies; these rows are shown for completeness. The exact value for $(1, 5)$ is 0.5785 (Section 5.2).

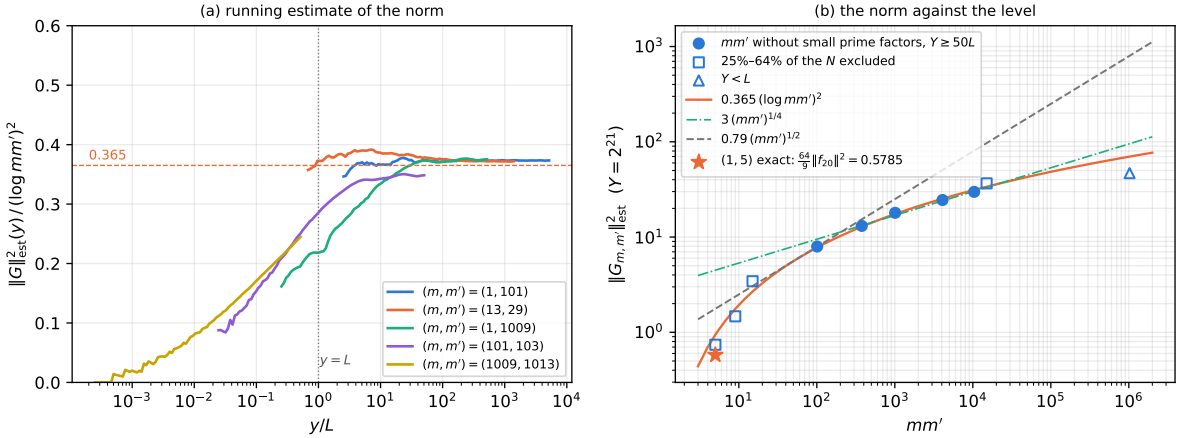



FIGURE 1. (a) The running estimate $\|G\|^2_{\mathrm{est}}(y)$, computed over the admissible $N \leq y$, divided by $(\log mm')^2$, against $y/L$ on a logarithmic scale, for five pairs $(m, m')$; the plateau for $y \gg L$ is the growth $Y^2$ of Theorem 1.2 and the Rankin–Selberg asymptotic of Remark 2.1, and its height is the constant 0.365 of Section 5.3 (dashed). (b) $\|G_{m,m'}\|^2_{\mathrm{est}}$ at $Y = 2^{21}$ against $mm'$, both axes logarithmic; filled circles: the five pairs with $mm'$ free of small prime factors and $Y \geq 50L$; open squares: pairs with 25%–64% of the $N$ excluded; triangle: $(1009, 1013)$, for which $Y < L$; star: the exact value at $(1, 5)$. The lines are $0.365(\log mm')^2$, the fit of Section 5.3, $3(mm')^{1/4}$, and $0.79(mm')^{1/2}$ (the growth allowed by Waibel's general bound, through the point $(1, 101)$). Script `plot_mean_square.py`; floating point.

5.2. **An exact check at the level 20.** The space $S_2(\Gamma_0(20))$ is one-dimensional, spanned by the newform $f_{20} = \eta(2z)^2\eta(10z)^2$ of the elliptic curve $20a1$: $y^2 = x^3 + x^2 + 4x + 4$, which is $X_0(20)$ itself (modular degree 1, Manin constant 1). Hence $G_{1,5} = c\, f_{20}$ with $c = a_G(1) = T(1, 5; 1) - M(1, 5; 1) = 4 - \frac{4}{3} = \frac{8}{3}$ (the Siegel product at $n = 1$ is $\frac{\pi^2}{5} \cdot \frac{25}{3\pi^2} \cdot \frac{4}{5}$), and $\|G_{1,5}\|^2 = \frac{64}{9}\|f_{20}\|^2$. The script `norm_level20.py` computes $\|f_{20}\|^2$ in two independent ways: from the Rankin–Selberg asymptotic $\sum_{n \leq y} a_f(n)^2/n \sim 16\pi^2\|f\|^2 y/\operatorname{vol}$, $\operatorname{vol}(\Gamma_0(20)\backslash\mathbb{H}) = 12\pi$, with the coefficients of the eta product up to $2^{20}$, which gives 0.08128, 0.08128, 0.08134 at $y = 2^{16}, 2^{18}, 2^{20}$; and from the period lattice $\Lambda$ of $20a1$, since $\varphi^*\omega = 2\pi i f(z)\mathrm{d}z$ gives $\|f\|^2 = \operatorname{covol}(\Lambda)/(4\pi^2)$ for a modular parametrisation of degree 1. The periods $\omega_1 = 2.824375\ldots$ (real) and $\omega_2 = 2.274165\ldots i$ of $y^2 = (x + 1)(x^2 + 4)$ are computed by quadrature, the lattice $\mathbb{Z}\omega_1 + \mathbb{Z}(\omega_1 + \omega_2)/2$ is identified as the period lattice by recomputing $g_2 = -44/3$ and $|g_3| = 296/27$ from it through $E_4$ and $E_6$, and its covolume $3.2115478\ldots$ gives $\|f_{20}\|^2 = 0.0813495\ldots$. The two values agree to four decimals. The script also checks that $T(1, 5; N) - M(1, 5; N)$ as computed by the pipeline of Part I equals $\frac{8}{3}a_f(N)$ on every admissible $N \leq 2^{20}$, to $10^{-9}$. This validates the Siegel main term, the FFT computation, the Rankin–Selberg normalisation and the volume used in the estimate $\|G\|^2_{\mathrm{est}}$. The exact value $\|G_{1,5}\|^2 = 0.5785$ is below the estimate 0.74 of Table 1, as expected: the excluded $N$ are either multiples of 5, where $|a_f(5n)| = |a_f(n)|$, or divisible by 16, where $a_f = 0$, so the admissible mean is biased upwards. With the main term made exact for every $N$ (Section 6.4), the estimate over all $N \leq 2^{21}$ is 0.5785, in agreement with the exact value to four decimals.

5.3. **The empirical law.** The last column of Table 1 shows $\|G\|^2_{\mathrm{est}}/(\log mm')^2$. For the five pairs with $10^2 \leq mm' \leq 1.1 \cdot 10^4$, $Y \geq 50L$ and $mm'$ free of small prime factors, it lies between

0.348 and 0.375, with mean 0.365, while the ratio to $(mm')^{1/2}$ falls from 0.79 to 0.29 and the ratio to $(mm')^{1/4}$ rises from 2.5 to 3.2 over the same range. Over a range of a factor 100 in $mm'$ the fits $(\log mm')^2$ and $(mm')^{1/4}$ are not separated by the data alone, but Theorem 1.1(i) excludes any power, and the heuristic of Part I, Remark 2.13, that $\|G\|^2$ is a sum over the cusps of $\sum_{n\ll mm'} |b(n)|^2/n$ with $|b(n)| \approx r(\alpha)r(\beta)$ on $O(1)$ pairs, predicts $(\log mm')^2$ with a constant of the order of $\frac{1}{4\pi}\cdot 2\cdot$(number of cusp classes with a large contribution) $\approx 0.3$–$0.5$. We therefore record

$$\|G_{m,m'}\|^2 \approx 0.37\,(\log mm')^2 \qquad (10^2 \le mm' \le 1.6\cdot 10^4),$$

and, for the root mean square of $(T-M)/\sqrt{N}$ over $N \le Y$, $Y \gg L$,

$$\mathrm{rms}^2 \approx \frac{16\pi^2\|G\|^2}{\mathrm{vol}(\Gamma_0(L)\backslash\mathbb{H})} \approx \frac{44\,(\log mm')^2}{mm'\prod_{p\mid 2mm'}(1+1/p)},$$

which reproduces the column "rms" of Table 3 of Part I (0.27 for $(101,103)$, 0.67 for $(1,1009)$). This corrects the remark in Section 5.3 of the first two versions of Part I that the root mean square "decreases like $(mm')^{-1/4}$" and that Waibel's bound "appears to be sharp": the decrease is like $(mm')^{-1/2}$ up to a logarithm, as Theorem 1.2 with Theorem 1.1(i) shows; version 3 of Part I states the corrected law. Section 6 shows what the "law" is: up to a few percent, the norm is the sum of the integrals of $|G|^2$ over the regions above the cusps, which are $\frac{1}{\pi}((\log m)^2 + (\log m')^2) + \frac{5}{\pi^2}\log m \log m' + \cdots$ with terms for the other divisors of $mm'$; the constant 0.37 is the average of this over the sample, and the estimates of Table 1 are themselves biased by the restriction to the admissible $N$ (Table 3 has the unbiased values).

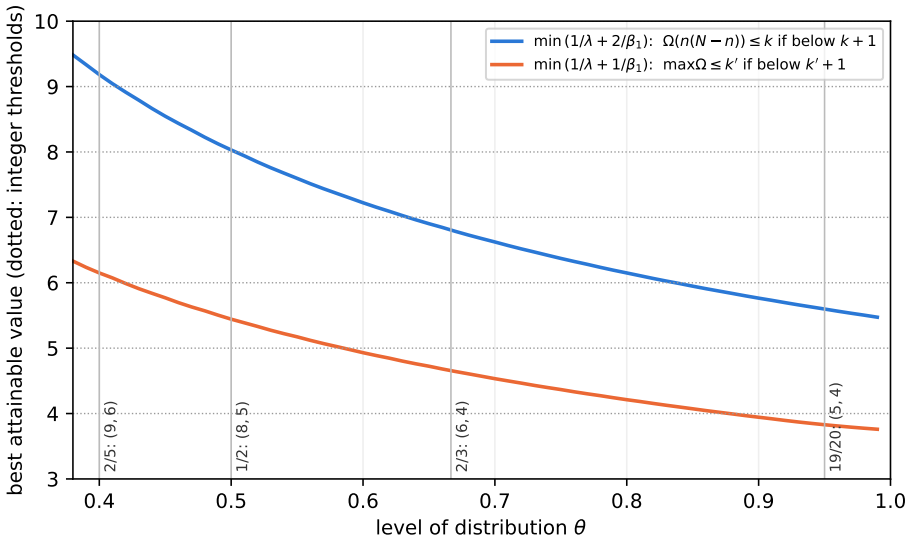



FIGURE 2. The smallest values of $1/\lambda + 2/\beta_1$ and of $1/\lambda + 1/\beta_1$ attainable with $L - \lambda R > 0$ in the sieve of Part I, as functions of the level of distribution $\theta$ (floating-point model of `modular_sieve_scan.py`, minimised over $\alpha$, $\beta_1$ with $\lambda = L/R$; script `plot_sieve_thresholds.py`). The bound $\Omega(n(N-n)) \le k$, resp. $\max\Omega \le k'$, holds when the blue, resp. orange, curve is below the dotted line $k+1$, resp. $k'+1$. The vertical lines mark $\theta = 2/5$ (Waibel's bound alone) and $1/2$ (Part I, every $N$), $2/3$ and $19/20$ (almost all $N$); the parameter sets at these four levels are certified.

5.4. **The supremum over $N$.** Conjecture 4.1 says that the supremum of $|a_G(N)|/\sqrt{N}$ over $N$ in a range is, up to $N^{\varepsilon}L^{\varepsilon}$, of the size of the root mean square $\approx 2.7(\log mm')(mm')^{-1/2}$ of Section 5.3. Table 2, extracted from the scans of Part I (`results/sup_error_scan_x21.txt`, `_x23.txt` there), gives the ratio of the supremum over all admissible $N \in (X/2, X]$ to the root mean square, for $X = 2^{21}$ (about $10^6$ values of $N$ per pair) and $X = 2^{23}$ (about $4 \cdot 10^6$). For the pairs with $L \le 6 \cdot 10^4$ the ratio lies between 5.4 and 9.8 and shows no trend in $mm'$; its mild growth from $X = 2^{21}$ to $2^{23}$ is the growth of the extreme value of a fixed heavy-tailed distribution with the sample size (the suprema are attained at $N$ with 16–192 divisors, in line with the factor $d(N)$ of Deligne's bound). The pair $(9, 1)$ is special: $S_2(\Gamma_0(36))$ is spanned by the newform of the curve $36a1$, $y^2 = x^3 + 1$, which has complex multiplication, so that $a_f(p) = 0$ for $p \equiv 2 \pmod 3$ and the distribution of $a_f(N)$ is far from the Gaussian bulk of Figure 3 of Part I. The pair $(1009, 1013)$ is in the range $X < 2L$, where neither the mean square nor the supremum has reached its asymptotic regime.

| $(m, m')$ | $mm'$ | rms | sup, $X = 2^{21}$ | sup /rms | sup, $X = 2^{23}$ | sup /rms |
|---|---|---|---|---|---|---|
| $(1, 5)$ | 5 | 1.76 | 14.7 | 8.4 | 17.3 | 9.8 |
| $(3, 5)$ | 15 | 1.90 | 10.5 | 5.5 | 12.2 | 6.4 |
| $(1, 101)$ | 101 | 1.40 | 11.8 | 8.4 | 12.1 | 8.6 |
| $(13, 29)$ | 377 | 0.89 | 5.9 | 6.6 | 6.8 | 7.7 |
| $(1, 1009)$ | 1009 | 0.67 | 4.4 | 6.5 | 5.3 | 8.0 |
| $(61, 67)$ | 4087 | 0.38 | 2.5 | 6.5 | 2.4 | 6.3 |
| $(101, 103)$ | 10403 | 0.27 | 1.9 | 7.3 | 1.8 | 6.7 |
| $(1, 15015)$ | 15015 | 0.18 | 1.0 | 5.7 | – | – |
| $(9, 1)$ | 9 | 1.76 | 28.7 | 16.3 | 33.8 | 19.2 |
| $(1009, 1013)$ | 1022117 | 0.036 | 0.61 | 17 | 0.55 | 14 |

TABLE 2. The supremum of $|T(m, m'; N) - M(m, m'; N)|/\sqrt{N}$ over the admissible $N \in (X/2, X]$ of Part I, Section 5, and its ratio to the root mean square over the same $N$ (the rms is the same to two decimals for both $X$). The last two pairs are the degenerate cases discussed in the text.

## 6. A lower bound for the norm: the regions above the cusps

Throughout this section $m, m'$ are coprime odd positive integers, $L = 4mm'$, $\chi = \chi_{-4}$, and for $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{Q})$ and $k \in \{1, 2\}$ we write $(f|_k\gamma)(z) = (\det\gamma)^{k/2}(cz + d)^{-k} f(\gamma z)$; thus scalar matrices act trivially, $f(mz) = m^{-k/2}(f|_k B_m)(z)$ with $B_m = \left(\begin{smallmatrix} m & 0 \\ 0 & 1 \end{smallmatrix}\right)$, and $(fg)|_2\gamma = (f|_1\gamma)(g|_1\gamma)$. We use $\theta^2 \in M_1(\Gamma_0(4), \chi)$, i.e. $\theta^2|_1\gamma = \chi(d)\theta^2$ for $\gamma \in \Gamma_0(4)$, and $\theta(-1/(4z)) = \sqrt{-2iz}\,\theta(z)$, which gives $\theta^2|_1 \left(\begin{smallmatrix} 0 & -1 \\ 4 & 0 \end{smallmatrix}\right) = -i\theta^2$.

6.1. **The Atkin–Lehner cusps.** For $Q \mid L$ with $(Q, L/Q) = 1$, an Atkin–Lehner matrix is $W_Q = \left(\begin{smallmatrix} Qa & b \\ Lc & Qd \end{smallmatrix}\right)$ with $a, b, c, d \in \mathbb{Z}$ and $\det W_Q = Q$ [1]. It normalises $\Gamma_0(L)$; hence for $f \in M_2(\Gamma_0(L))$ the form $f|_2 W_Q$ lies in $M_2(\Gamma_0(L))$ and does not depend on the choice of $W_Q$ (two choices differ by an element of $\Gamma_0(L)$ on either side), $f \mapsto f|_2 W_Q$ is an isometry for the Petersson product which maps cusp forms to cusp forms, and it therefore preserves the orthogonal decomposition of $M_2(\Gamma_0(L))$ into the Eisenstein and the cuspidal subspace.

**Lemma 6.1.** *Let $Q \in \{m, 4m\}$. Then $\Theta_{m,m'}|_2 W_Q = \varepsilon_Q\, \Theta_{1,mm'}$ with $|\varepsilon_Q| = 1$, and consequently $E_{m,m'}|_2 W_Q = \varepsilon_Q E_{1,mm'}$ and $G_{m,m'}|_2 W_Q = \varepsilon_Q G_{1,mm'}$. The same holds with $m$ and $m'$ interchanged.*

*Proof.* Let first $Q = m$ and $W_m = \left(\begin{smallmatrix} ma & b \\ Lc & md \end{smallmatrix}\right)$, $mad - 4m'bc = 1$. Then $B_m W_m = m \left(\begin{smallmatrix} ma & b \\ 4m'c & d \end{smallmatrix}\right)$ with the second factor in $\Gamma_0(4)$, so $\theta^2(m\cdot)|_1 W_m = m^{-1/2}\theta^2|_1(B_m W_m) = m^{-1/2}\chi(d)\theta^2$; and $W_m = \left(\begin{smallmatrix} a & b \\ 4m'c & md \end{smallmatrix}\right) B_m$ with the first factor in $\Gamma_0(4m')$, so $\theta^2(m'\cdot)|_1 W_m = \chi(md)\,\theta^2(m'\cdot)|_1 B_m = \chi(md)\, m^{1/2}\theta^2(mm'\cdot)$. Multiplying, $\Theta_{m,m'}|_2 W_m = \chi(m)\,\theta^2(z)\theta^2(mm'z) = \chi(m)\Theta_{1,mm'}$.

Let now $Q = 4m$ and $W_{4m} = \left(\begin{smallmatrix} 4ma & b \\ Lc & 4md \end{smallmatrix}\right)$, $4mad - m'bc = 1$. Then $B_m W_{4m} = m \left(\begin{smallmatrix} 4ma & b \\ 4m'c & 4d \end{smallmatrix}\right)$, and the second factor is an Atkin–Lehner matrix $W_4$ for $\Gamma_0(4)$; every such matrix is $\left(\begin{smallmatrix} 0 & -1 \\ 4 & 0 \end{smallmatrix}\right)$ times an element of $\Gamma_0(4)$, so $\theta^2|_1 W_4 = \pm i\theta^2$ and $\theta^2(m\cdot)|_1 W_{4m} = \pm i\, m^{-1/2}\theta^2$. Also $W_{4m} = W_4' B_m$ with $W_4' = \left(\begin{smallmatrix} 4a & b \\ 4m'c & 4md \end{smallmatrix}\right)$ an Atkin–Lehner matrix for $\Gamma_0(4m')$, and $B_{m'} W_4' = \left(\begin{smallmatrix} 4a & m'b \\ 4c & 4md \end{smallmatrix}\right) B_{m'}$ with an Atkin–Lehner matrix for $\Gamma_0(4)$ in front, so $\theta^2(m'\cdot)|_1 W_4' = m'^{-1/2}(\pm i\theta^2)|_1 B_{m'} = \pm i\,\theta^2(m'\cdot)$ and $\theta^2(m'\cdot)|_1 W_{4m} = \pm i\, m^{1/2}\theta^2(mm'\cdot)$. Multiplying, $\Theta_{m,m'}|_2 W_{4m} = \pm\Theta_{1,mm'}$. The assertions on $E$ and $G$ follow because $|_2 W_Q$ preserves the orthogonal decomposition and $\Theta_{1,mm'} = E_{1,mm'} + G_{1,mm'}$ is that decomposition for $\Theta_{1,mm'}$ (Siegel's theorem applied to $Q_{1,mm'}$). □

We now use the formula of Part I, Lemma 2.8 (Waibel's [11, (22)]): with $\mu = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(L)]$, $\tilde\mu = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma(L)]$, $\tilde\mu/\mu = L\varphi(L)$, $\mathcal{F}$ the standard fundamental domain of $\mathrm{SL}_2(\mathbb{Z})$, $T = \{\left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right) : 0 \le t \le L-1\}$ and $\varrho_{a,c} = \left(\begin{smallmatrix} a & * \\ c & * \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$,

$$\|G\|^2 = \frac{\mu}{\tilde\mu} \sum_{c \ (L)} \sum_{\substack{a \ (L) \\ (a,c,L)=1}} I(\varrho_{a,c}), \qquad I(\varrho) = \int_{\bigcup_{t\in T} t\mathcal{F}} \big|(G|_2\varrho)(z)\big|^2 \,\mathrm{d}x\,\mathrm{d}y, \tag{10}$$

for $G \in S_2(\Gamma_0(L))$, where $(a,c)$ runs over coprime representatives of the residue classes. Every term is non-negative, and $\bigcup_{t\in T} t\mathcal{F} \supseteq \{-\frac12 \le x \le L - \frac12,\ y \ge 1\}$.

**Lemma 6.2.** *Let $Q \in \{m, 4m, m', 4m'\}$ and $a(n) = a_{G_{1,mm'}}(n)$. The terms of (10) for $G = G_{m,m'}$ with $(c, L) = L/Q$ contribute in total at least*

$$\frac{1}{4\pi} \sum_{n\ge1} \frac{|a(n)|^2}{n}\, e^{-4\pi n/Q}.$$

*Proof.* Let $(c, L) = L/Q$, $c = (L/Q)c_1$ with $(c_1, Q) = 1$, and $(a, c) = 1$; then $(a, L/Q) = 1$, so $(aQ, (L/Q)c_1) = 1$ and there are $b, d \in \mathbb{Z}$ with $aQd - b(L/Q)c_1 = 1$. The matrix $\varrho = \left(\begin{smallmatrix} a & b \\ (L/Q)c_1 & Qd \end{smallmatrix}\right)$ is in $\mathrm{SL}_2(\mathbb{Z})$, has the prescribed first column, and $\varrho B_Q = \left(\begin{smallmatrix} Qa & b \\ Lc_1 & Qd \end{smallmatrix}\right)$ is an Atkin–Lehner matrix $W_Q$. Hence, by Lemma 6.1 and $B_Q^{-1} = Q^{-1}\left(\begin{smallmatrix} 1 & 0 \\ 0 & Q \end{smallmatrix}\right)$,

$$G_{m,m'}|_2\varrho = (G_{m,m'}|_2 W_Q)|_2 B_Q^{-1} = \varepsilon_Q\, G_{1,mm'}|_2 \left(\begin{smallmatrix} 1 & 0 \\ 0 & Q \end{smallmatrix}\right) = \frac{\varepsilon_Q}{Q}\, G_{1,mm'}\Big(\frac{z}{Q}\Big) = \frac{\varepsilon_Q}{Q} \sum_{n\ge1} a(n)\, e\Big(\frac{nz}{Q}\Big).$$

This function has period $Q$ in $x$, and $Q \mid L$, so by Parseval over $-\frac12 \le x \le L - \frac12$,

$$I(\varrho) \ \ge\ \int_1^\infty \int_{-1/2}^{L-1/2} |G|_2\varrho|^2 \,\mathrm{d}x\,\mathrm{d}y = \frac{L}{Q^2} \sum_{n\ge1} |a(n)|^2 \int_1^\infty e^{-4\pi n y/Q}\,\mathrm{d}y = \frac{L}{4\pi Q} \sum_{n\ge1} \frac{|a(n)|^2}{n} e^{-4\pi n/Q}.$$

The number of pairs $(a, c)$ modulo $L$ with $(c, L) = L/Q$ and $(a, c, L) = 1$, i.e. $(a, L/Q) = 1$, is $\varphi(Q)\cdot\varphi(L/Q)\,Q = \varphi(L)Q$, and $\mu/\tilde\mu = 1/(L\varphi(L))$. Multiplying gives the assertion. □

*Proof of Theorem 1.5(i).* Since $m, m'$ are odd and coprime with $mm' > 1$, we have $m \ne m'$, $m \ne 4m'$ and $m' \ne 4m$, so the four values $L/Q$, $Q \in \{m, 4m, m', 4m'\}$, are distinct and the four sets of pairs $(a, c)$ in Lemma 6.2 are disjoint. (When $m = 1$ the values $Q = 1$ and $Q = 4$ occur: $W_1 \in \Gamma_0(L)$, the class $(c, L) = L$ is the cusp $\infty$ itself and the expansion is that of $G_{1,m'}$; Lemmas 6.1 and 6.2 cover these cases as stated, since they use only the Atkin–Lehner matrices

and Parseval's identity, and the four cusps are then all the Atkin–Lehner cusps of $\Gamma_0(4m')$.) As all terms of (10) are non-negative, (i) follows by adding the four lower bounds. □

6.2. **The sum $\sum r(n)^2/n$.**

**Lemma 6.3.** *For $Y \ge 1$,*

$$S(Y) := \sum_{n\ge1} \frac{r(n)^2}{n}\, e^{-n/Y} = 2(\log Y)^2 + c_1 \log Y + c_2 + O(Y^{-1/2}),$$

$$c_1 = 4\gamma + 8\frac{L'}{L}(1,\chi) - 8\frac{\zeta'}{\zeta}(2) + \frac{4}{3}\log 2 = 9.758\ldots,$$

*with an absolute constant $c_2$ ($c_2 = 6.12\ldots$ numerically); in particular $S(Y) \ge 2(\log Y)^2 - C$ for $Y \ge 1$. Moreover $\sum_{n\ge1} r(n)^2 e^{-n/X} = 4X\log X + O(X)$, so that $\sum_{n\le X} r(n)^2 \ll X\log 2X$, for $X \ge 1$.*

*Proof.* Since $r(n)/4 = \sum_{d|n}\chi(d)$ is multiplicative with $r(p^k)/4 = k+1$, $\frac12(1+(-1)^k)$, 1 for $p \equiv 1,3 \pmod 4$ and $p = 2$, the series $\sum_k (r(p^k)/4)^2 x^k$ equals $(1+x)(1-x)^{-3}$, $(1-x^2)^{-1}$, $(1-x)^{-1}$ respectively, and

$$D(s) = \sum_{n\ge1}\frac{r(n)^2}{n^s} = \frac{16\,\zeta(s)^2 L(s,\chi)^2}{(1+2^{-s})\,\zeta(2s)} \qquad (\operatorname{Re} s > 1).$$

By Mellin inversion, $S(Y) = \frac{1}{2\pi i}\int_{(1)} \Gamma(w) Y^w D(1+w)\,\mathrm{d}w$. We move the contour to $\operatorname{Re} w = -\frac12$: there $\zeta(2+2w) \ne 0$ and $1/\zeta(2+2w) \ll \log(|w|+2)$, $\zeta(1+w)$ and $L(1+w,\chi)$ are $\ll (|w|+2)^{1/4}$, and $\Gamma(w)$ decays exponentially, so the integral on the new line is $O(Y^{-1/2})$. The only pole crossed is $w = 0$, of order three. With $\Gamma(w) = w^{-1}(1-\gamma w + O(w^2))$, $Y^w = 1 + w\log Y + \frac12 w^2(\log Y)^2 + O(w^3)$ and $D(1+w) = 4w^{-2}(1 + a_1 w + O(w^2))$, where $16\cdot\frac23\cdot L(1,\chi)^2/\zeta(2) = 16\cdot\frac23\cdot\frac{\pi^2}{16}\cdot\frac{6}{\pi^2} = 4$ and $a_1 = 2\gamma + 2\frac{L'}{L}(1,\chi) - 2\frac{\zeta'}{\zeta}(2) + \frac13\log 2$, the residue is $2(\log Y)^2 + 4(a_1 - \gamma)\log Y + c_2$. The numerical value uses $L'(1,\chi)/L(1,\chi) = \gamma + 2\log 2 + 3\log\pi - 4\log\Gamma(\frac14) = 0.2456\ldots$ and $\zeta'(2)/\zeta(2) = -0.5700\ldots$. Since $c_1 > 0$, $S(Y) \ge 2(\log Y)^2 + c_2 - O(1)$ for $Y \ge 1$. The last assertion is the same argument for $\frac{1}{2\pi i}\int \Gamma(w) X^w D(w)\mathrm{d}w$ with the contour moved to $\operatorname{Re} w = \frac34$, where only the double pole at $w = 1$ is crossed, and $\sum_{n\le X} r(n)^2 \le e\sum_n r(n)^2 e^{-n/X}$. □

6.3. **The main term for small $n$.**

**Lemma 6.4.** *Let $Q = Q_{m,m'}$ and $n \ge 1$. Then $\beta_2(n) \le 2$; $\beta_p(n) \le \sigma_{-1}(p^{v_p(n)})$ for $p \nmid 2mm'$; and $\beta_p(n) \le 7\,(\min(v_p(n), v_p(mm')) + 1)$ for $p \mid mm'$, while $\beta_p(n) \le 1 + 1/p$ for $p \mid mm'$, $p \nmid n$. Consequently, for $Y \ge 1$,*

$$\sum_{n\ge1} \frac{r(n)\,M(m,m';n)}{n}\, e^{-n/Y} \;\ll\; \frac{Y}{mm'}\,\log(2Y)\,(\log\log 3mm')^C$$

*with an absolute constant $C$.*

*Proof.* For a prime $p$, $a \ge 1$ and $t \in \mathbb{Z}$ let $N_2(t;p^a) = \#\{(x,y) \bmod p^a :\ x^2 + y^2 \equiv t\}$. We claim

$$N_2(t;p^a) \le 2\,(\min(v_p(t), a) + 1)\,p^a \quad (p \text{ odd}), \qquad N_2(t;2^a) \le 2^{a+1}. \tag{11}$$

For odd $p$, the solutions with $(x,y) \not\equiv (0,0) \pmod p$ are non-singular and lift from $p$ to $p^a$ with multiplicity $p^{a-1}$; modulo $p$ there are $p - \chi(p) \le 2p$ of them if $p \nmid t$ and $2(p-1)$ or 0 if $p \mid t$. The solutions with $x \equiv y \equiv 0 \pmod p$ exist only if $p^2 \mid t$ and are $p^2 N_2(t/p^2; p^{a-2})$ in number for $a \ge 2$. Hence $N_2(t;p^a) \le 2p^a + p^2 N_2(t/p^2; p^{a-2})$, and (11) follows by induction on $a$ ($N_2(t;p) \le 2p$, $N_2(t;1) = 1$). For $p = 2$ and $a \ge 3$: if $t \equiv 1 \pmod 4$, exactly one of $x, y$ is

odd, and since the odd squares modulo $2^a$ are the residues $\equiv 1 \pmod 8$, each taken four times, while $y^2 \equiv 0$ or $4 \pmod 8$ according as $4 \mid y$ or not, there are $2 \cdot 4 \cdot 2^{a-2} = 2^{a+1}$ solutions; if $t \equiv 2 \pmod 8$ both are odd and there are $4 \cdot 4 \cdot 2^{a-3} = 2^{a+1}$ solutions; if $t \equiv 3 \pmod 4$ or $t \equiv 6 \pmod 8$ there are none; and if $4 \mid t$ both are even and $N_2(t; 2^a) = 4N_2(t/4; 2^{a-2})$, so (11) follows by induction from $N_2(t; 2^a) \le 2^{a+1}$ for $a \le 2$, which one checks by hand.

Now $\beta_2(n) = \lim_a 2^{-3a} \sum_{(z,w) \bmod 2^a} N_2\big((n - m'(z^2+w^2))\bar m; 2^a\big) \le \lim_a 2^{-3a} \cdot 2^{2a} \cdot 2^{a+1} = 2$, where $\bar m$ is an inverse of $m$ modulo $2^a$. For $p \nmid 2mm'$ the formula $\beta_p(n) = (1-p^{-2})\sigma_{-1}(p^{v_p(n)})$ is (6) of Part I. Let $p \mid mm'$, say $p^e \parallel m$ with $e \ge 1$, $p \nmid m'$, and $p^k \parallel n$. By (11), with $X = x^2 + y^2$,

$$\beta_p(n) = \lim_a p^{-3a} \sum_{(x,y) \bmod p^a} N_2\big((n - mX)\bar m'; p^a\big) \le 2\Big(1 + \sum_{i \ge 1} \mathrm{Prob}_{(x,y)}\big(p^i \mid n - mX\big)\Big),$$

the probability referring to $(x, y)$ uniform modulo $p^a$ with $a \ge i$. For $i \le \min(k, e)$ the probability is 1. If $k < e$ it is 0 for $i > k$. If $k \ge e$ and $i > e$, then $p^i \mid n - mX$ if and only if $X \equiv n_1 \pmod{p^{i-e}}$ with $n_1 = (n/p^e)(m/p^e)^{-1}$, which has probability $N_2(n_1; p^{i-e})p^{-2(i-e)} \le 2(i-e+1)p^{-(i-e)}$ by (11), and $\sum_{j \ge 1} 2(j+1)p^{-j} \le \frac{5}{2}$ for $p \ge 3$. Hence $\beta_p(n) \le 2(\min(k,e) + \frac{7}{2}) \le 7(\min(k,e)+1)$; for $k = 0$ the exact value $1 - \chi(p)/p$ is (7) of Part I.

For the sum, write $M(m, m'; n) = \pi^2 n\, \beta(n)/(mm')$ with $\beta(n) = \prod_p \beta_p(n)$, so that the sum is $\frac{\pi^2}{mm'} \sum_n r(n)\beta(n)e^{-n/Y}$, and $\beta(n) \le 2\,\sigma_{-1}(n) \prod_{p \mid mm'}(1 + \frac{1}{p}) \prod_{p \mid (n, mm')} 7(v_p(n)+1)$. Write $n = gk$ with $g$ composed of the primes of $mm'$ and $(k, mm') = 1$; then $r(n) \le 4d(g)d(k)$, $\sigma_{-1}(n) \le 2^{\omega(g)}\sigma_{-1}(k)$, and $\sum_{k \le x} d(k)\sigma_{-1}(k) \ll x \log 2x$ (write $\sigma_{-1}(k) = \sum_{u \mid k} 1/u$ and sum over $u$ first). Therefore

$$\sum_n r(n)\beta(n)e^{-n/Y} \ll \prod_{p \mid mm'}\Big(1 + \frac{1}{p}\Big) \sum_g \frac{7^{\omega(g)}\, 2^{\omega(g)}\, d(g)^2}{g}\, Y \log(2Y) \ll Y \log(2Y) \prod_{p \mid mm'}\Big(1 + \frac{C}{p}\Big),$$

since $\sum_{j \ge 1} 14(j+1)^2 p^{-j} \le C/p$ for $p \ge 3$, and $\prod_{p \mid mm'}(1 + C/p) \le \exp(C \sum_{p \mid mm'} 1/p) \ll (\log\log 3mm')^C$. □

*Proof of Theorem 1.5(ii).* Fix $Q \in \{m, 4m, m', 4m'\}$ with $Q \ge 4\pi$ and put $Y = Q/(4\pi) \ge 1$; note $Y \le \max(m, m')/\pi \le mm'/\pi$. Keep only $n < mm'$ in the inner sum of (i), where $a(n) = r(n) - M(1, mm'; n)$ and $M \ge 0$, so that $|a(n)|^2 \ge r(n)^2 - 2r(n)M(1, mm'; n)$:

$$\sum_{n \ge 1} \frac{|a(n)|^2}{n} e^{-n/Y} \ge S(Y) - \sum_{n \ge mm'} \frac{r(n)^2}{n} e^{-n/Y} - 2 \sum_{n \ge 1} \frac{r(n) M(1, mm'; n)}{n} e^{-n/Y}.$$

By Lemma 6.3, $S(Y) \ge 2(\log Y)^2 - C$, and the tail is at most

$$\sum_{j \ge 0} e^{-\pi 2^j} \frac{1}{2^j mm'} \sum_{n < 2^{j+1} mm'} r(n)^2 \ll \sum_{j \ge 0} e^{-\pi 2^j} \log(2^{j+2} mm') \ll \log(2mm').$$

By Lemma 6.4 with $(m, m')$ replaced by $(1, mm')$, the last sum is at most a constant times $(Y/mm')\log(2Y)(\log\log 3mm')^C$, which is at most $\log(2mm')(\log\log 3mm')^C$ because $Y/mm' \le 1/\min(m, m') \le 1$. Summing over the four $Q$ and using

$$2\big(\log(m/4\pi)\big)^2 + 2\big(\log(m/\pi)\big)^2 \ge 4(\log m)^2 - 4\log(4\pi^2)\log m$$

(and the same for $m'$), while the terms with $Q < 4\pi$, if any, are dropped and the corresponding $(\log m)^2 \le (\log 12)^2$ is absorbed in the constant, (i) gives

$$\|G_{m,m'}\|^2 \ge \frac{1}{\pi}\Big((\log m)^2 + (\log m')^2\Big) - \frac{\log(4\pi^2)}{\pi} \log(mm') - C \log(2mm')(\log\log 3mm')^C,$$

which is (ii); the second inequality in (ii) is $(\log m)^2 + (\log m')^2 \ge \frac{1}{2}(\log mm')^2$. □

6.4. **All cusps, numerically.** For square-free $m, m'$ the computation of Lemma 6.2 can be carried out at every cusp. The cusps of $\Gamma_0(L)$ are then in bijection with the divisors $d$ of $L$ (one cusp of denominator $d$ for each $d$, since $(d, L/d) \le 2$), and we write $d = d_2\delta$ with $d_2 \mid 4$, $\delta \mid mm'$, $\delta_m = (\delta, m)$, $\delta_{m'} = (\delta, m')$, and

$$A = \delta_m \frac{m'}{\delta_{m'}}, \qquad B = \delta_{m'} \frac{m}{\delta_m}, \qquad AB = mm', \quad (A,B) = 1.$$

The cusps with $d_2 \in \{1, 4\}$ are the images of $\infty$ under the Atkin–Lehner involutions $W_{L/d}$, their widths are $L/d = 4mm'/\delta$ and $mm'/\delta$, and the proof of Lemma 6.1, applied prime by prime, gives $G_{m,m'}|_2 W_{L/d} = \varepsilon\, G_{A,B}$ with $|\varepsilon| = 1$ (the odd part of $L/d$ moves the primes of $\delta_m$ from the first block to the second and those of $\delta_{m'}$ from the second to the first; the 2-part contributes a sign). The two cusps of Theorem 1.5 with the pair $(1, mm')$ are $\delta \in \{m, m'\}$. At the cusp with $d_2 = 2$, the function $\theta^2$ is transformed into $\theta_2^2$, $\theta_2(z) = \sum_{x \text{ odd}} e(x^2 z/4)$ (from $\theta^2|_1 \left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right) = \theta_2^2$, which follows from the transformation formulae of $\theta_3, \theta_4, \theta_2$ under $z \mapsto -1/(4z)$), and $\Theta_{m,m'}$ into $(mm'/\delta)^{-1} \sum_n T^{\mathrm{odd}}(A, B; n) e(\delta n z/(4mm'))$, where $T^{\mathrm{odd}}$ counts the representations of $n$ by $Q_{A,B}$ with all four coordinates odd; the Eisenstein part of this expansion is given by Siegel's theorem for the coset $(1 + 2\mathbb{Z})^4$ of $2\mathbb{Z}^4$, i.e. by $M^{\mathrm{odd}}(A, B; n) = \frac{\pi^2 n}{AB} \cdot \frac{1}{2}[n \equiv 2(A + B)\ (8)] \prod_{p>2} \beta_p(n)$ (for $(A, B) = (1, 1)$ this is Jacobi's formula $16\sigma(n/4)$ for the number of representations of $n \equiv 4 \pmod 8$ as a sum of four odd squares). Integrating $|G|^2$ over the region above each cusp as in Lemma 6.2 gives

$$\|G_{m,m'}\|^2 \ \ge\ \Lambda(m, m') := \sum_{\delta \mid mm'} \Big[ \frac{1}{4\pi} \sum_{n \ge 1} \frac{|a_{G_{A,B}}(n)|^2}{n} \big( e^{-\pi\delta n/mm'} + e^{-4\pi\delta n/mm'} \big) + \frac{1}{\pi} \sum_{n \ge 1} \frac{|T^{\mathrm{odd}}(A, B; n) - M^{\mathrm{odd}}(A, B; n)|^2}{n}\, e^{-\pi\delta n/mm'} \Big], \tag{12}$$

where $A, B$ depend on $\delta$ as above. The regions above all the cusps have total area equal to the sum of the widths, which is $\mu = L \prod_{p \mid L} (1 + 1/p)$, i.e. the fraction $3/\pi = 0.955$ of $\mathrm{vol}(\Gamma_0(L) \backslash \mathbb{H})$; the rest of the fundamental domain lies below height 1 at every cusp.

The script `cusp_lower_bound.py` evaluates (12). It needs the main terms $M(A, B; n)$ and $M^{\mathrm{odd}}(A, B; n)$ for *every* $n$, not only the admissible ones: the module `local_densities.py` computes the densities at the primes $p \mid AB$ dividing $n$ by the recursion $\beta_p(Q_{A,B}; pn') = 2(1 - \frac{1}{p})[p \equiv 1\ (4)] + \frac{1}{p}\beta_p(Q_{A/p,pB}; n')$ for $p \mid A$ (the non-singular solutions have $(z, w) \not\equiv 0$, the singular ones reduce to the form $Q_{A/p,pB}$), which gives $\beta_p(n) = (1 + \frac{1}{p})p^{-k}$ for $p \equiv 3$ and $2 - (1 + \frac{1}{p})p^{-k}$ for $p \equiv 1 \pmod 4$, $k = v_p(n)$ (the recursion closes because $Q_{A/p,pB}$ is again a form of the family, with $p$ moved to the other block, and $\beta_p = 1 - \chi(p)/p$ when $p \nmid n$); the 2-adic density by sorting the solutions by the parities of $(x, y)$ and $(z, w)$, which gives $\beta_2(n) = \frac{1}{4}\beta_2(n/4) + \frac{1}{2}[A \equiv B\ (4)][v_2(n) = 2] + [A \not\equiv B\ (4)](1 + \frac{1}{2}[v_2(n) \ge 3])$ for $v_2(n) \ge 2$; and $\beta_2^{\mathrm{odd}}(n) = \frac{1}{2}[n \equiv 2(A + B)\ (8)]$. These formulae are checked exactly: at the levels 4 and 12, where there are no cusp forms, $T = M$ and $T^{\mathrm{odd}} = M^{\mathrm{odd}}$ for every $n \le 2^{17}$ to $10^{-9}$, and at the level 20, $T - M = \frac{8}{3} a_{f_{20}}(n)$ for every $n \le 2^{17}$ (Section 5.2 had this only for the admissible $n$), while the mean square of $T^{\mathrm{odd}} - M^{\mathrm{odd}}$ grows like $Y^2$ as for a cusp form. With the exact main term the Rankin–Selberg estimate of Section 5 can be taken over all $n \le Y$; this is the column $\|G\|^2$ of Table 3, and for $(1, 5)$ it is 0.5785, the exact value of Section 5.2.

| $(m,m')$ | $mm'$ | $\|G\|^2$ | Tab. 1 | $\Lambda_{\mathrm{AL}}$ | $\Lambda_{\mathrm{odd}}$ | $\Lambda$ | $\Lambda/\|G\|^2$ | Thm 1.5(i) | $\frac{1}{\pi}(\log^2 m+\log^2 m')$ |
|---|---|---|---|---|---|---|---|---|---|
| $(1,5)$ | 5 | 0.5785 | 0.74 | 0.496 | 0.046 | 0.543 | 0.938 | 0.496 | 0.82 |
| $(3,5)$ | 15 | 2.058 | 3.44 | 1.799 | 0.136 | 1.936 | 0.940 | 1.061 | 1.21 |
| $(3,7)$ | 21 | 2.481 | 4.02 | 2.186 | 0.167 | 2.352 | 0.948 | 1.183 | 1.59 |
| $(5,7)$ | 35 | 4.109 | 4.82 | 3.555 | 0.314 | 3.869 | 0.942 | 2.154 | 2.03 |
| $(1,101)$ | 101 | 7.429 | 7.95 | 7.147 | 0.100 | 7.247 | 0.976 | 7.147 | 6.78 |
| $(13,29)$ | 377 | 12.423 | 13.09 | 10.795 | 1.033 | 11.828 | 0.952 | 6.639 | 5.70 |
| $(15,77)$ | 1155 | 21.779 | 27.93 | 19.620 | 1.340 | 20.960 | 0.962 | 9.596 | 8.34 |
| $(1,1009)$ | 1009 | 16.937 | 17.96 | 16.572 | 0.106 | 16.678 | 0.985 | 16.572 | 15.23 |
| $(61,67)$ | 4087 | 24.704 | 24.47 | 21.411 | 2.000 | 23.411 | 0.948 | 12.875 | 11.01 |
| $(101,103)$ | 10403 | 30.385 | 29.81 | 26.470 | 2.447 | 28.917 | 0.952 | 15.847 | 13.62 |
| $(105,143)$ | 15015 | 47.140 | 49.50 | 40.622 | 3.186 | 43.807 | 0.929 | 17.079 | 14.73 |
| $(1,15015)^*$ | 15015 | 43.233 | 43.54 | 44.610 | 0.230 | 44.839 | 1.037 | 30.818 | 29.44 |
| $(1009,1013)^*$ | 1022117 | 57.730 | 61.23 | 58.545 | 5.271 | 63.816 | 1.105 | 34.469 | 30.47 |

TABLE 3. The lower bound (12) and its parts (`results/cusp_lower_bound.txt`, `_large.txt`): $\Lambda_{\mathrm{AL}}$ is the sum over the Atkin–Lehner cusps ($d_2\in\{1,4\}$), $\Lambda_{\mathrm{odd}}$ the sum over the cusps with $d_2=2$, $\Lambda$ their sum, "Thm 1.5(i)" the right-hand side of Theorem 1.5(i) (the four cusps with $\delta\in\{m,m'\}$), and the last column its leading term. $\|G\|^2$ is the Rankin–Selberg estimate over all $n\le 2^{21}$ with the exact main term ($2^{24}$ for the two pairs marked $^*$), and "Tab. 1" the estimate over the admissible $n$ of Section 5 (for $(3,7)$, $(5,7)$, $(15,77)$, $(105,143)$ computed here in the same way). For the two pairs marked $^*$ the estimate has not converged ($Y/L=280$ and 4; Table 1 has $S_2(Y)/S_2(Y/2)=4.36$ and 4.56 for them), and the lower bound is the better information. Floating point.

Table 3 shows the following.

(a) For the ten pairs with $Y\ge 50L$, the regions above the cusps carry between 93.8% and 98.5% of the norm (for $(105,143)$, with $Y/L=35$, 92.9%), about their share 95.5% of the area: the function $y^2|G|^2$ has no excess mass in the remaining 4.5% of the fundamental domain, near the elliptic points and in the "corners" between the cusps. The lower bound (12) is thus nearly an equality, and an asymptotic formula for $\|G_{m,m'}\|^2$ is a question about the lower part of the fundamental domain only.

(b) The four cusps of Theorem 1.5(i) alone give 98% for $m=1$, where they are all the Atkin–Lehner cusps, and about half for $mm'=pq$; the rest comes from the pair $(m,m')$ itself at the cusps with $\delta=1$ (widths $4mm'$ and $mm'$, and the cusp of denominator 2): for $(13,29)$, $4.16+1.03$ out of 12.42.

(c) For the cusps with $\delta=1$, $T(m,m';n)$, $n\ll mm'$, is a sum of at most $1+n/mm'$ products $r(\alpha)r(\beta)$ with $m\alpha+m'\beta=n$; replacing $r(\alpha)^2, r(\beta)^2$ by their mean values $4\log\alpha$, $4\log\beta$ (Lemma 6.3) gives for their Atkin–Lehner part

$$\frac{16}{4\pi}\iint_{u,v>0}\frac{\log(m'u)\log(mv)}{u+v}\bigl(e^{-\pi(u+v)}+e^{-4\pi(u+v)}\bigr)\mathrm{d}u\,\mathrm{d}v=\frac{5}{\pi^2}\log m\log m'+O(\log mm'),$$

since $\iint e^{-c(u+v)}(u+v)^{-1}\mathrm{d}u\,\mathrm{d}v=1/c$: 4.38, 8.76, 10.84 against the computed 4.16, 8.54, 10.62 for $(13,29)$, $(61,67)$, $(101,103)$.

(d) The leading term $\frac{1}{\pi}((\log m)^2+(\log m')^2)$ of Theorem 1.5(i) is 85–95% of the right-hand side of (i) for $mm'\ge 100$; the difference is the term $c_1\log Y+c_2$ of Lemma 6.3 at the four cusps, which is still large at $Y\approx 10^2$–$10^3$.

(e) When $mm'$ has several prime factors, the other divisors contribute: for $m=1$ and $\delta\mid m'$ the cusps with the pair $(m'/\delta,\delta)$ and width $m'/\delta$ see $T(m'/\delta,\delta;n)=r(n/\delta)$ for $n<m'/\delta$,

hence contribute about $\frac{1}{\pi\delta}\big(\log\frac{m'}{\delta^2}\big)_+^2$: for $(1,15015)$ the divisors $\delta=3,5,7,11,13$ give 6.72, 2.88, 1.78, 0.82, 0.57 (against 5.84, 2.61, 1.49, 0.67, 0.49 from the formula), which is why this pair has the largest ratio $\|G\|^2/(\log mm')^2$ in Table 1.

In short, $\|G_{m,m'}\|^2$ is, to within a few percent, $\frac{1}{\pi}((\log m)^2+(\log m')^2)+\frac{5}{\pi^2}\log m\log m'$ plus the contribution of the cusp of denominator 2 and of the other divisors of $mm'$, plus lower-order terms; its ratio to $(\log mm')^2$ tends to $1/\pi$ for $(1,m')$ and to $\frac{1}{2\pi}+\frac{5}{4\pi^2}+\cdots$ for $m\approx m'$ prime, with lower-order terms that are still substantial at $mm'\approx 10^4$. The constant 0.37 of Section 5.3 is the average of these ratios over the sample.

## 7. The two-dimensional sieve: $(7,5)$ for every $N$

Remark 3.3 observed that the product $n(N-n)$ is a sieve problem of dimension 2, with the level of distribution $N^{1/2-\varepsilon}$ of Part I, Theorem 3.2, and that the two-dimensional sieve of Diamond, Halberstam and Richert has the sieving limit $\beta_2=4.2665$ against $2+2\sqrt{e}=5.297$ for the vector sieve of Part I. Here we carry this out.

7.1. **The sieve functions of dimension two.** We recall the definitions [7], [6, Chapters 6 and 9]. For $\kappa\ge1$ the Ankeny–Onishi function $\sigma_\kappa$ is the continuous solution of

$$\sigma_\kappa(s)=\frac{s^\kappa}{(2e^\gamma)^\kappa\,\Gamma(\kappa+1)}\quad(0<s\le2),\qquad\big(s^{-\kappa}\sigma_\kappa(s)\big)'=-\kappa s^{-\kappa-1}\sigma_\kappa(s-2)\quad(s>2),$$

and the sieve functions $F_\kappa, f_\kappa$ are the solutions of

$$\begin{aligned}&F_\kappa(s)=\frac{1}{\sigma_\kappa(s)}\quad(0<s\le\alpha_\kappa),&&\big(s^\kappa F_\kappa(s)\big)'=\kappa s^{\kappa-1}f_\kappa(s-1)\quad(s>\alpha_\kappa),\\&f_\kappa(s)=0\quad(0<s\le\beta_\kappa),&&\big(s^\kappa f_\kappa(s)\big)'=\kappa s^{\kappa-1}F_\kappa(s-1)\quad(s>\beta_\kappa),\end{aligned}\tag{13}$$

where $\alpha_\kappa,\beta_\kappa$ are the constants for which $F_\kappa(s),f_\kappa(s)\to1$; $\alpha_1=\beta_1=2$ and $F_1,f_1$ are the linear sieve functions, and $\alpha_\kappa>\beta_\kappa>2$ for $\kappa>1$. $F_\kappa$ decreases and $f_\kappa$ increases, $F_\kappa=1+O(e^{-s})$, $f_\kappa=1-O(e^{-s})$, and both are Lipschitz: $|F_\kappa(s_1)-F_\kappa(s_2)|,|f_\kappa(s_1)-f_\kappa(s_2)|\le\kappa|s_2-s_1|/(s_1\sigma_\kappa(1))$ for $1\le s_1<s_2$ [6, Lemma 6.2]. The sieve theorem [6, Theorem 9.1] (in the form quoted in [8]) states, for a sequence $\mathcal{A}$ with $|\mathcal{A}_d|=X\omega(d)/d+R_d$, $\omega$ multiplicative with $\sum_{w\le p<z}\omega(p)\log p/p\le\kappa\log(z/w)+O(1)$, $V(z)=\prod_{p<z}(1-\omega(p)/p)$ and $2\le z\le y$,

$$\begin{aligned}S(\mathcal{A},z)&\le XV(z)\Big(F_\kappa\Big(\frac{\log y}{\log z}\Big)+E\Big)+2\sum_{\substack{m\mid P(z)\\ m<y}}4^{\omega(m)}|R_m|,\\S(\mathcal{A},z)&\ge XV(z)\Big(f_\kappa\Big(\frac{\log y}{\log z}\Big)-E\Big)-2\sum_{\substack{m\mid P(z)\\ m<y}}4^{\omega(m)}|R_m|,\end{aligned}\tag{14}$$

with $E=O\big((\log\log y)^2(\log y)^{-1/(2\kappa+2)}\big)$.

The constants $\alpha_\kappa,\beta_\kappa$ are determined by two conditions, which we formulate with the adjoint functions $p_\kappa,q_\kappa$, the solutions of $s^\kappa p'(s)=-\kappa(s+1)^{\kappa-1}p(s+1)$ with $p(s)\sim s^{-\kappa}$ and of $s^\kappa q'(s)=\kappa(s+1)^{\kappa-1}q(s+1)$ with $q(s)\sim s^\kappa$ (Booker and Browning [2] use the equivalent $(up)'=\kappa(p(u)-p(u+1))$, $(uq)'=\kappa(q(u)+q(u+1))$). For $P=F_\kappa+f_\kappa$ and $Q=F_\kappa-f_\kappa$ the

quantities

$$\Phi_P(s) = s^\kappa P(s)p(s) + \kappa\int_{s-1}^{s} (u+1)^{\kappa-1}P(u)p(u+1)\,\mathrm{d}u,$$
$$\Phi_Q(s) = s^\kappa Q(s)q(s) - \kappa\int_{s-1}^{s} (u+1)^{\kappa-1}Q(u)q(u+1)\,\mathrm{d}u$$

are constant for $s \ge \max(\alpha_\kappa,\beta_\kappa)+1$, by the differential-delay equations, and $P\to 2$, $Q\to 0$ if and only if $\Phi_P = 2$ and $\Phi_Q = 0$ (a solution $Q$ of its equation which is not the sieve solution decays only like $s^{-2\kappa}$, and then $\Phi_Q \neq 0$). The script `dhr_sieve.py` solves (13) for given $(\alpha,\beta)$ on a grid of step $1/2000$, computes $p, q$ by backward integration from their asymptotic series, and solves $\Phi_P = 2$, $\Phi_Q = 0$ for $(\alpha,\beta)$. For $\kappa = 1$ it returns $\alpha = \beta = 2$ and reproduces $F_1(3) = 2e^\gamma/3$, $f_1(4) = 2e^\gamma\log 3/4$ and $f_1(5/2)$ to six decimals. For $\kappa = 2$ it returns

$$\alpha_2 = 5.357727, \qquad \beta_2 = 4.266450,$$

which agree to $10^{-8}$ with the values $\alpha_2 = 5.35772744559\ldots$, $\beta_2 = 4.26645028414\ldots$ computed in ball arithmetic by Booker and Browning [2] (truncated at 20 decimals there), and with $\beta_2 = 4.2665$ of [7]; also $F_2(s) = 2(2e^\gamma)^2/s^2 = 25.378/s^2$ on $(0,2]$. Refining the grid to $1/4000$ changes $\alpha_2,\beta_2$ by $3\cdot 10^{-8}$ and the values of $F_2, f_2$ below by less than $10^{-7}$. Some values:

| $s$ | 2 | 3 | 4 | 4.5 | 5 | 6 | 8 | 10 |
|---|---|---|---|---|---|---|---|---|
| $F_2(s)$ | 6.3444 | 2.9164 | 1.8364 | 1.5704 | 1.3928 | 1.1136 | 1.0027 | 1.00002 |
| $f_2(s)$ | 0 | 0 | 0 | 0.2403 | 0.5790 | 0.8844 | 0.9973 | 0.99998 |

7.2. **The weighted sieve.** We keep the setting of Part I, Section 4: $N \equiv 2 \pmod 4$ large, $\mathcal{A} = \{n \le N :\ (n,N) = 1\}$ with the weights $w(n) = r(n)r(N-n)$, $X_N = \sum_{n\in\mathcal{A}} w(n)$ and, for square-free $d$ with $(d,2N) = 1$,

$$\sum_{\substack{n\in\mathcal{A}\\ d\mid n(N-n)}} w(n) = X_N\prod_{p\mid d} 2h(p) + R_d, \qquad R_d = \sum_{d_1d_2=d} R(d_1,d_2;N),$$

since $(n,N) = 1$ forces $d \mid n(N-n)$ to split uniquely as $d_1 \mid n$, $d_2 \mid N-n$ (Remark 3.3); here $h(p) = (2p-1)/(p(p+1))$ for $p\equiv 1$ and $h(q) = 1/(q(q-1))$ for $q\equiv 3 \pmod 4$, as in Part I, (18). Therefore $\sum_{d\le D} d_3(d)|R_d|$ is bounded by the sum of Part I, Theorem 3.2, which is $\ll N^{1-\varepsilon/3}$ for $D = N^{1/2-\varepsilon}$, and the same holds for the sums over $d \le D/p$ needed for the sequences $\mathcal{A}_p = \{n\in\mathcal{A} :\ p \mid n(N-n)\}$. The density $\omega(p) = 2h(p)p$ satisfies $\sum_{p<z} 2h(p) = 2\log\log z + O(1)$, so the sieve has dimension $\kappa = 2$ in the sense of (14). Let $0 < a < b < \frac12$, $\lambda > 0$, $z = N^a$, $y = N^b$, and consider

$$W = \sum_{\substack{n\in\mathcal{A}\\ (n(N-n),P(z))=1}} w(n)\Big(1 - \lambda\sum_{\substack{z\le p<y\\ p\mid n(N-n)}}\Big(1-\frac{\log p}{\log y}\Big)\Big) = S(\mathcal{A},z) - \lambda\sum_{z\le p<y}\Big(1-\frac{\log p}{\log y}\Big)S(\mathcal{A}_p,z).$$

By (14) with the lower function for $\mathcal{A}$ at the level $N^{1/2-\varepsilon}$ and the upper function for $\mathcal{A}_p$ at the level $N^{1/2-\varepsilon}/p$, and with the passage from the sum over $p$ to an integral as in Part I, Section 4,

$$W \ \ge\ X_N V(z)\big(L_2 - \lambda R_2 - o(1)\big), \qquad L_2 = f_2\Big(\frac{1}{2a}\Big), \qquad R_2 = 2\int_a^b\Big(1-\frac{t}{b}\Big)F_2\Big(\frac{1/2-t}{a}\Big)\frac{\mathrm{d}t}{t} \tag{15}$$

(the $-\varepsilon$ in the level is absorbed by the Lipschitz continuity of $f_2, F_2$). On the other hand, as in Part I, Lemma 4.2, the $n$ with $p^2 \mid n(N-n)$ for some $p \ge z$ contribute $o(X_NV(z))$, and for the others the weight is positive only if $\Omega(n(N-n)) < 1/\lambda + 2/b$ and $\max(\Omega(n),\Omega(N-n)) < 1/\lambda + 1/b$, because $\sum_{p\mid n}(1-\log p/\log y) \ge \Omega(n) - \log n/\log y \ge \Omega(n) - 1/b$ and likewise for $N-n$, while the terms with $p \ge y$ are $\le 0$ and all prime factors are $\ge z$. Hence, exactly as in the proof of

Theorem 1.1 of Part I, if $L_2-\lambda R_2>0$, then for every large $N\equiv 2 \pmod 4$ the number of $n\in\mathcal{A}$ with $n$, $N-n$ products of distinct primes $\equiv 1 \pmod 4$ exceeding $N^a$, $\Omega(n(N-n))\le k$ and $\max(\Omega(n),\Omega(N-n))\le k'$, where $k+1\ge 1/\lambda+2/b$ and $k'+1\ge 1/\lambda+1/b$, is at least

$$\big(L_2-\lambda R_2-o(1)\big)\,2^{-(k+4)}\cdot 8\,\mathfrak{S}_r(N)\,c_{2h}\,\frac{N}{a^2(\log N)^2}, \tag{16}$$

with $c_{2h}=1.66485$ as there (the factor $2^{-(k+4)}$ is $1/\max w(n)$, since $r(n)=4\cdot 2^{\omega(n)}$ for such $n$).

**Theorem 7.1.** *For every sufficiently large $N\equiv 2 \pmod 4$,*

$$\#\big\{1\le n<N:\ n=x^2+y^2,\ N-n=z^2+w^2,\ \Omega(n(N-n))\le 7,\ \max(\Omega(n),\Omega(N-n))\le 5\big\} \ge\ 0.25\,\mathfrak{S}_r(N)\frac{N}{(\log N)^2},$$

*the integers counted being coprime to $N$ and products of distinct primes $\equiv 1 \pmod 4$ exceeding $N^{1/18}$.*

**Lemma 7.2.**

$$f_2(9)\ \ge\ 0.9995, \qquad R_2^*:=2\int_{1/18}^{2/5}\Big(1-\frac{5t}{2}\Big)F_2(9-18t)\,\frac{\mathrm{d}t}{t}=2\int_{1.8}^{8}\frac{s-1.8}{7.2\,(9-s)}\,F_2(s)\,\mathrm{d}s\ \le\ 2.625. \tag{17}$$

*Proof.* This is a computation in interval arithmetic, in the manner of Part I, Appendix A; the script is `certify_dhr.py` and its output is in the repository. Its inputs are: the enclosures

$$\alpha_2=5.35772744559446184227\ldots, \qquad \beta_2=4.26645028414864191641\ldots$$

of [2], computed there in ball arithmetic and correctly truncated at the 20th decimal (adding $10^{-20}$ gives upper bounds); the closed form $s^{-2}\sigma_2(s)=A_2(4-2\log(s/2)-8/s+4/s^2)$ on $[2,4]$, $A_2=1/(2(2e^{\gamma})^2)$, which follows from the delay equation of $\sigma_2$, and on $[4,\alpha_2]$ one more integration of that equation, enclosed cell by cell (the factor $u^{-3}(u-2)^2$ increases for $u<6$ and $s^{-2}\sigma_2(s)$ decreases); and the monotonicity of $F_2$ (decreasing) and $f_2$ (increasing) [6, Lemma 6.2], which bounds every cell integral of (13) by its endpoint values. Closed forms are evaluated in 80-bit interval arithmetic and the recursions in IEEE doubles with an outward rounding step after each operation, on the grid of step $1/20000$. The enclosures obtained are $f_2(9)\in[0.999711,\ 0.999738]$ and $R_2^*\le 2.623903$; the floating-point values 0.99972 and 2.6238 of Section 7.1 lie inside. At $s=4.5,5,6,8$ the certified enclosures of $F_2, f_2$ have widths below $3\cdot 10^{-5}$ and contain the values of the table above. □

*Proof of Theorem 7.1.* Take $a=1/18$, $b=2/5$, $\lambda=1/3$ in (15)–(16). Then $1/\lambda+2/b=8$ and $1/\lambda+1/b=5.5$, so $k=7$ and $k'=5$ are admissible, $L_2=f_2(9)$, $R_2=R_2^*$, and $L_2-\lambda R_2\ge 0.9995-2.625/3=0.1245$ by Lemma 7.2; then (16) gives the constant $0.1245\cdot 2^{-11}\cdot 8\cdot 1.66485\cdot 18^2=0.2623>0.25$. The remainder sums in (14) carry the weight $4^{\omega(m)}$ in place of the $d_3(de)$ of Part I, Theorem 3.2; the proof there bounds any such weight of size $N^{o(1)}$ by $N^{\eta}$ (its condition (F1)), so the theorem holds with this weight as well, and the sums over $m<N^{1/2-\varepsilon}/p$ for $\mathcal{A}_p$ are sub-sums of the same sum. □

Theorem 7.1 improves the pair $(8,5)$ of Part I, Theorem 1.1, to $(7,5)$ from the same analytic input, the level $1/2$, and it is unconditional: the sieve functions enter only through Lemma 7.2, whose inputs are the certified constants of [2] and the monotonicity of $F_2, f_2$.

7.3. **Comparison with the vector sieve.** Table 4 compares, at the three levels of distribution that occur in these papers, the smallest attainable values of $1/\lambda + 2/b$ and $1/\lambda + 1/b$ for the two sieves (floating point; the vector sieve values are those of Figure 2), and the largest margin $L_2 - \lambda R_2$ for the pairs of interest. At the level $1/2$ the vector sieve attains 8.03 and 5.44, just missing $(7, \cdot)$, while the two-dimensional sieve attains 7.56 and 5.07: $(7, 5)$ is obtained with margin 0.13, and $(7, 4)$ is missed by 0.03. At $2/3$ (the level that Proposition 4.2 would give for every $N$ from a saving $(mm')^{-1/4}$) both sieves give $(6, 4)$, the two-dimensional one with margin 0.196 against 0.037; at $19/20$ the two-dimensional sieve gives $(5, 3)$ with margin 0.24, where the vector sieve reached $(5, 3)$ only at the edge of its parameter range (Section 3).

| $\theta$ | vector sieve | | dimension 2 | | margin $L_2 - \lambda R_2$ for the pair, at $(a, b, \lambda)$ | |
|---|---|---|---|---|---|---|
| | $1/\lambda + 2/b$ | $1/\lambda + 1/b$ | $1/\lambda + 2/b$ | $1/\lambda + 1/b$ | | |
| $1/2$ | 8.03 | 5.44 | 7.56 | 5.07 | $(7,5)$: $+0.132$ $(0.054, 0.42, 0.309)$ | $(7,4)$: $-0.030$ |
| $2/3$ | 6.79 | 4.64 | 6.37 | 4.50 | $(6,4)$: $+0.196$ $(0.076, 0.50, 1/3)$ | $(5,4)$: $-0.157$ |
| $19/20$ | 5.60 | 3.83 | 5.21 | 3.75 | $(5,3)$: $+0.238$ $(0.124, 0.50, 1/2)$ | $(4,3)$: $-0.099$ |

TABLE 4. The two sieves for $n(N - n)$ (`results/dhr_sieve.txt`, `results/sieve_thresholds.txt`): the bound $\Omega(n(N-n)) \le k$, resp. $\max \Omega \le k'$, requires $1/\lambda + 2/b \le k + 1$, resp. $1/\lambda + 1/b \le k' + 1$, with $L - \lambda R > 0$. The vector sieve values at $2/3$ are those of Figure 2 at $\theta = 0.67$. Floating point.

## 8. Beyond the level 1/2: the sum over the moduli

We end with the question that the sieve route leaves open for every $N$: the level of distribution beyond $1/2$. At the level $\theta$, the remainder of the sieve (the vector sieve of Part I as well as the two-dimensional sieve of Section 7) is a sum

$$\sum_{\substack{de \le N^\theta \\ (d,e)=1,\ (de,2N)=1}} \lambda_d \lambda'_e \, R(d, e; N) \tag{18}$$

with sieve weights $|\lambda_d|, |\lambda'_e| \le 1$ supported on square-free integers, and by (4) each $R(d, e; N)$ is a bounded combination of coefficients $a_{G_{d',e'}}(N')$ with $d'e' \asymp de$ (up to divisors of $N$) and $N' \in \{N, N/2\}$. Part I bounds (18) by $\sum |R(d, e; N)|$ and each term by $N^{1/2+\varepsilon}$ (Theorem 1.1(ii)), which gives $N^{1/2+\varepsilon} N^\theta$ and hence $\theta < 1/2$; Proposition 4.2 shows that any power saving $(de)^{-\eta}$ in the individual bound raises the level. The other possibility is cancellation between the terms: $R(d, e; N)$ changes sign with $(d, e)$, and the sieve weights are sign-changing too.

8.1. **The structure of the sum.** Since $\sum_{d|n,\ e|N-n} r(n) r(N - n) = T(d, e; N)$ for coprime $d, e$, the sum (18) is, before the Möbius inversion of $(n, N) = 1$,

$$\sum_{d,e} \lambda_d \lambda'_e \big(T(d, e; N) - M(d, e; N)\big) = \sum_{d,e} \lambda_d \lambda'_e \Big( \sum_{d\alpha + e\beta = N} r(\alpha) r(\beta) - M(d, e; N) \Big),$$

a weighted count of the representations of $N$ by the family of quaternary forms $Q_{d,e}$. Writing $n = d\alpha$ and $N - n = e\beta$, the main sum is

$$\sum_{e} \lambda'_e \sum_{\substack{d,\alpha \\ d\alpha \equiv N \ (e)}} \lambda_d \, r(\alpha) \, r\Big(\frac{N - d\alpha}{e}\Big), \tag{19}$$

the distribution of the products $d\alpha$, weighted by $\lambda_d r(\alpha)$, in the residue class $N$ modulo $e$, with the additional weight $r((N-d\alpha)/e)$ attached to the quotient. Without the last factor this is the setting

of the dispersion method of Bombieri, Friedlander and Iwaniec [3]: sums $\sum_q \lambda_q \sum_{mn\equiv a\ (q)} \alpha_m\beta_n$ with well-factorable $\lambda$, for which the level $4/7$ is known for suitable $\alpha,\beta$ from the Deshouillers–Iwaniec bounds for sums of Kloosterman sums. The factor $r((N-d\alpha)/e) = 4\sum_{k\mid(N-d\alpha)/e}\chi(k)$ introduces a third modulus: (19) becomes $4\sum_{e,k}\lambda'_e\chi(k)\sum_{d\alpha\equiv N\ (ek)}\lambda_d r(\alpha)$, the distribution of $d\alpha$ in progressions to the moduli $ek$, where $k$ runs up to $N/(de)$, so that the moduli $ek$ range up to $N$. This is the shape of the Titchmarsh divisor problem, $\sum_q\sum_{mn\equiv a\ (q)}$ over all $q\le x$, which the dispersion method does handle (Fouvry, Bombieri–Friedlander–Iwaniec), with the difference that here both $d\alpha$ and the quotient carry the arithmetic weight $r$ and the modulus $e$ carries a sieve weight. We expect that the dispersion method, applied to (19) with the moduli $e$ in a range $N^{\theta_e}$ and the bilinear structure of $\lambda_d r(\alpha)$, gives a level $\theta>1/2$ for the signed sum (18) and with it an improvement of Theorem 7.1 for every $N$; we have not carried this out. The spectral version of the same question is the sum over the levels $4de$ of the inner products $\langle G_{d,e}, P_N^{(4de)}\rangle$ of Section 4.

8.2. **The signs of $a_{G_{d,e}}(N)$, numerically.** Whether cancellation of this kind is present can be measured. The script `modulus_sum_scan.py` computes, for a fixed $N\equiv 2 \pmod 4$ and all coprime odd square-free pairs $(d,e)$ with $de\le N^{3/4}$, the coefficients $a_{d,e}=a_{G_{d,e}}(N)=T(d,e;N)-M(d,e;N)$ exactly (the main term for every $N$ by the formulae of Section 6.4), and compares the sums

$$A=\sum|a_{d,e}|,\qquad Q=\Big(\sum a_{d,e}^2\Big)^{1/2},$$
$$S_1=\sum a_{d,e},\qquad S_\mu=\sum\mu(de)\,a_{d,e},\qquad S_{\mathrm{rnd}}=\sum\pm a_{d,e}$$

(all sums over the pairs with $de\le N^\theta$), where $\pm$ are independent random signs; $|S|/Q\approx 1$ is square-root cancellation and $A/Q$ is the maximum. Table 5 shows the result.

| $N$ | $\theta$ | pairs | $A/N$ | $Q/N$ | $S_1/Q$ | $S_\mu/Q$ | $S_{\mathrm{rnd}}/Q$ | $A/Q$ |
|---|---|---|---|---|---|---|---|---|
| $2000002=2\cdot101\cdot9901$ | 0.50 | 2149 | 0.82 | 0.024 | −0.43 | 0.66 | 1.93 | 35 |
| | 0.60 | 10367 | 2.43 | 0.033 | −0.75 | 0.55 | −0.05 | 73 |
| | 0.70 | 49699 | 6.78 | 0.044 | 1.43 | −0.76 | 0.39 | 154 |
| | 0.75 | 108127 | 10.93 | 0.049 | 2.01 | −1.30 | 0.69 | 221 |
| $2000006=2\cdot1000003$ | 0.50 | 2149 | 0.85 | 0.025 | 2.15 | 0.38 | 0.29 | 35 |
| | 0.60 | 10367 | 2.49 | 0.034 | 2.34 | −0.11 | 1.50 | 74 |
| | 0.70 | 49699 | 6.83 | 0.044 | 0.40 | −0.20 | 1.82 | 154 |
| | 0.75 | 108131 | 11.01 | 0.050 | 1.74 | −0.75 | 2.88 | 221 |
| $2000018=2\cdot293\cdot3413$ | 0.50 | 2149 | 0.86 | 0.025 | −1.87 | −1.34 | 0.63 | 34 |
| | 0.60 | 10367 | 2.49 | 0.034 | −1.78 | −0.54 | 1.09 | 73 |
| | 0.70 | 49699 | 6.77 | 0.045 | −1.13 | 1.04 | −0.08 | 152 |
| | 0.75 | 108131 | 10.90 | 0.050 | 0.71 | 1.29 | 0.89 | 218 |
| $3000002=2\cdot557\cdot2693$ | 0.50 | 2677 | 0.78 | 0.021 | 0.51 | 0.37 | 1.08 | 38 |
| | 0.60 | 13515 | 2.35 | 0.029 | 1.05 | −0.25 | 0.72 | 82 |
| | 0.70 | 67357 | 6.63 | 0.038 | 2.18 | −0.97 | 1.52 | 177 |
| | 0.75 | 149721 | 10.81 | 0.042 | 3.28 | −0.06 | 1.08 | 257 |

TABLE 5. Sums of $a_{G_{d,e}}(N)$ over the coprime odd square-free pairs with $de\le N^\theta$ (`results/modulus_sum_scan.txt`). Floating point.

For all three weightings the sums are of the size $Q$, the square root of the sum of squares, as for independent random signs ($|S_\mu|/Q\le 1.3$, $|S_{\mathrm{rnd}}|/Q\le 2.9$, $|S_1|/Q\le 3.3$ in the table): the coefficients $a_{G_{d,e}}(N)$ for different $(d,e)$ show no sign correlation with each other or with $\mu(de)$, and the signed sums are smaller than $\sum|a_{d,e}|$ by the factor $A/Q$, which grows with $\theta$

(from 35 at $\theta = 1/2$ to 220 at $3/4$). The sum $A$ itself is already larger than $N$ for $\theta \geq 0.55$ at these $N$, which is why the proofs need $N$ large; the signed sums are $0.02$–$0.05\,N$. Since $|a_{d,e}| \approx N^{1/2}(de)^{-1/2}\log(de)$ on average (Theorem 1.2 and Section 5), $Q \approx N^{1/2}(\log N)^{3/2}$ for every $\theta$, while $A \approx N^{1/2+\theta/2}$; both are $o(N)$ for $\theta < 1$ if Conjecture 4.1 holds, but only the cancellation visible here is available without it. The data are, of course, no substitute for a proof: they say that the sum (18) behaves as the dispersion method would predict, not how to prove it.

## Data and code

The scripts `mean_square_check.py`, `norm_level20.py`, `certify_modular.py` (arguments `1/2`, `2/3`, `19/20`), `plot_mean_square.py` and `plot_sieve_thresholds.py` (Figures 1 and 2), `local_densities.py` and `cusp_lower_bound.py` (Table 3), `dhr_sieve.py` (Section 7, output `results/dhr_sieve.txt`), `certify_dhr.py` (Lemma 7.2, output `results/certify_dhr_output.txt`), `modulus_sum_scan.py` (Table 5, output `results/modulus_sum_scan.txt`), the modules of Part I they use (`sup_error_scan.py`, `modular_sieve_scan.py`, `sieve_core.py`) and the outputs `results/mean_square_check_x21.txt`, `results/norm_level20.txt`, `results/certify_modular_output_twothirds.txt`, `results/certify_modular_output_nineteen.txt`, `results/sieve_thresholds.txt`, `results/cusp_lower_bound.txt`, `results/cusp_lower_bound_large.txt` are in the repository https://github.com/Ruqing1963/two-squares-mean-square; the code and data of Part I are at https://github.com/Ruqing1963/two-squares-theta-level. The certification of the sieve parameters uses directed rounding as described in Part I, Appendix A; the computations of Section 5 are in floating point and are evidence, not proof.

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GUT Geoservice Inc., Montreal, Canada

*Email address*: `ruqing@hotmail.com`